Calibeating Prediction-Powered Inference

Anonymous Author(s)
Affiliation
Address
email

Abstract

We study semisupervised mean estimation with a small labeled sample, a large unlabeled sample, and an available black-box prediction model that may be miscalibrated. This setting arises in randomized trials, biobanks, and other data-rich studies where gold-standard outcomes are expensive but surrogate predictions are readily available. Standard prediction-powered estimators such as augmented inverse-probability weighting (AIPW) [81] remain valid under prediction-model misspecification, but they can leave efficiency on the table when the prediction score is poorly calibrated.

We propose a simple post-hoc calibration correction: using the labeled sample, we recalibrate the prediction score and then average the calibrated predictions over the pooled sample. This procedure, which we call *Calibrated Prediction-Powered Inference*, requires no retraining of the original model and can improve efficiency relative to AIPW based on the original uncalibrated score. We study both linear and isotonic calibration. For isotonic calibration, we establish a first-order *calibeating* guarantee: the calibrated estimator can improve on the original score and on simpler post-processing rules, and no further post-processing of the fitted isotonic score yields additional first-order improvement. For linear calibration, we show first-order equivalence to PPI++ and hence the same asymptotic efficiency.

We also clarify that recent PPI methods are particular AIPW-based procedures within a broader semiparametric regression-adjustment framework. In simulations and benchmark experiments, our calibrated estimators often improve on PPI and remain competitive with, or outperform, AIPW and PPI++.

1 Introduction

We study semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem: rich covariate information is available for many individuals, but the outcome of interest is observed only for a smaller subset. Such settings arise when gold-standard outcomes are expensive to collect but prediction scores are readily available, for example in randomized trials with prognostic models trained on historical data [96], biobank analyses using phenotype predictors built from EHR or omics measurements [48, 128], and computational biology pipelines that supplement slow or costly experimental measurements with model-based predictions [1, 130]. The goal is to use the larger unlabeled sample to improve estimation of a population mean while accounting for prediction error. This setting has long been studied in missing-data and semiparametric statistics, where methods such as AIPW, TMLE, and related debiased estimators allow flexible machine-learning predictors to be used with valid inference [81, 114, 116, 115, 22]. More recently, related ideas have reappeared in the machine-learning literature under the name prediction-powered inference [1, 2, 130]. Closely related questions also

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

arise in covariate adjustment for randomized trials, where machine-learning predictors trained on auxiliary data can improve precision in small samples [41, 90, 67, 65, 96, 26, 51].

A standard approach is to use the labeled sample to debias a plug-in estimator based on a prediction model. If $m(X)$ is a black-box prediction score, the plug-in estimator averages $m(X)$ over the pooled covariate sample, while the labeled sample is used to estimate the residual correction term $\mathbb{E}\{Y-m(X)\}$. The augmented inverse-probability weighted (AIPW) estimator [81] averages $m(X)$ over both labeled and unlabeled samples and then adds the average residual over the labeled sample, whereas the PPI estimator [1] averages $m(X)$ only over the unlabeled covariates before applying the same residual correction. PPI can be less efficient because it discards labeled-covariate information. Thus, while these methods protect against bias from imperfect prediction models, they typically treat the prediction score itself as fixed. In practice, however, useful black-box prediction scores are often systematically too high, too low, or too extreme. Such misalignment may arise because the model was trained on a different population, or because of temporal drift, distribution shift, or regularization bias.

This motivates post-hoc calibration of the prediction score. Rather than correcting only the marginal bias $\mathbb{E}\{Y-m(X)\}$, one can use the labeled sample to transform the score so that its numerical values better align with the outcomes they are meant to predict [126, 127, 71]. An ideal target is a calibrated score $m_n^\star(X)$ satisfying $\mathbb{E}[Y \mid m_n^\star(X)] = m_n^\star(X)$. Under this condition, $\mathbb{E}[m_n^\star(X)] = \mathbb{E}[Y]$, so the corresponding plug-in estimator is automatically unbiased for the population mean. This is attractive because a useful prediction model may already be available, while a relatively small labeled sample can recalibrate its output without retraining [86]. Classical post-processing methods include Platt scaling [76], linear calibration [66], histogram binning [126], and isotonic regression [127]. Such calibration is especially relevant for modern machine-learning predictors, which are often informative yet systematically miscalibrated [7, 35, 122].

Our proposal is therefore simple: fit a black-box regression model, calibrate it on the labeled sample, and average the calibrated predictions over the pooled covariate sample. We call this workflow *Calibrated Prediction-Powered Inference*. This remains a simple "predict, calibrate, and average" procedure, but we show that it also admits an exact debiased interpretation. In particular, the resulting estimator is exactly equal to the AIPW estimator based on the calibrated score, with the residual correction encoded by the calibration step itself. Because calibration can improve the score, it can also improve efficiency relative to AIPW based on the uncalibrated model. Calibration therefore converts a simple plug-in rule into a debiased estimator with attractive efficiency properties.

Contributions. We make three main contributions. First, we introduce a general *Calibrated Prediction-Powered Inference* framework for semisupervised mean estimation, covering familiar post-hoc methods such as linear and isotonic calibration. Second, we show that calibration provides three gains: it can improve the prediction score and therefore precision; it makes the pooled plug-in estimator automatically debiased because it is exactly equal to AIPW with the calibrated score; and it can produce a fitted score that cannot be improved by further post-processing within the relevant class, a form of *calibeating* in the sense of Foster and Hart [31]. Third, we develop asymptotic theory for these estimators. In particular, isotonic-calibrated PPI gives a lightweight default with valid inference, efficiency gains under miscalibration, and no further first-order gain from post-processing, while linearly calibrated PPI is first-order equivalent to PPI++ and related to classical prognostic-score regression adjustment.

1.1 Related work

Although PPI has been presented as a recent development, the underlying problem and many of its estimators are special cases of a broader literature on semiparametric debiasing and regression adjustment, including work on missing data, causal inference, and debiased machine learning.

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes [91], and semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures [81, 82, 80, 114, 101, 10]. In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations and the efficiency framework used throughout this paper.

Debiased machine learning and targeted learning. A large body of work combines this semiparametric framework with machine-learning-based nuisance estimation under the labels of targeted learning and debiased machine learning [116, 115, 22, 24]. In the present setting, these methods already imply that flexible black-box predictors can be used with valid inference once the missingness probability is known, which is exactly the lens through which we interpret score-based semisupervised estimators.

Semisupervised inference and PPI. Related work studies semisupervised or surrogate-assisted estimation when the primary outcome is observed only on a limited subset [73, 74, 19, 18, 21, 50, 53]. More recent machine-learning papers popularized the label *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting [1, 2, 130]. These papers helped popularize the setting, but methodologically PPI estimators are best understood as instances or variants of existing semiparametric procedures. For example, PPI++ is AIPW with empirical-efficiency maximization [90, 114]. MEC [58] instead uses generalized entropy weights to balance the prediction score, which is closer to the balancing-weight representation of our linearly calibrated estimator. Closely related questions have long been studied in randomized trials [41, 84, 67, 68, 96, 47]. Here we emphasize the corresponding regression-adjustment view, implemented by ordinary linear regression of Y on the score.

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is statistically equivalent to randomized experiments and related sampling problems once the missingness indicator is reinterpreted as a treatment or sampling indicator [92, 114, 30, 70]. This equivalence allows tools from covariate adjustment, causal inference, and model-assisted estimation to transfer directly to the semisupervised setting.

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling, linear calibration, isotonic regression, and histogram binning improve predictive reliability without retraining the underlying model [76, 25, 37, 66, 126, 127, 71, 36]. Most closely related for our purposes is recent work connecting calibration to debiased estimation of statistical functionals [108, 105, 110].

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation

We observe two independent datasets,

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n \qquad \text{and} \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N.$$

The labeled sample consists of i.i.d. draws from a distribution P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled sample consists of i.i.d. draws from the distribution $P_{0,X}$ of X under P_0. Our target is the population mean $\psi_0 := \mathbb{E}_{P_0}[Y]$, and we write $\mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x]$.

For functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means over the labeled and unlabeled samples by

$$\mathbb{P}_n^L a := \frac{1}{n}\sum_{i=1}^n a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N}\sum_{j=1}^N b(\widetilde{X}_j).$$

For readability, we will sometimes write $\mathbb{P}_n^L\{a(X,Y)\}$ instead of $\mathbb{P}_n^L a$, and for functions of X alone we write $\mathbb{P}_n^L\{b(X)\} = n^{-1}\sum_{i=1}^n b(X_i)$. Let $M := n + N$ and $\rho_n := n/M$; pooled score averages are written as $\rho_n \mathbb{P}_n^L f + (1-\rho_n)\mathbb{P}_N^U f$.

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$, and the two samples are independent.

(ii) Y has finite second moment under P_0.

(iii) With $M := n + N$, the labeled fraction $\rho_n = n/M$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in (0,1)$.

2.2 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model [81]. It is helpful to view AIPW not as a single estimator, but as a family of estimators indexed by a score function $f : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such f, define

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1-\rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{1}$$

This estimator combines a plug-in term that imputes Y from $f(X)$ with a residual correction from the labeled sample. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for ψ_0. Thus, the choice of f affects efficiency through variance, but not validity. This makes AIPW a safe estimator [124] and a particularly simple instance of double robustness [5]; it is the semisupervised version of the broader covariate-adjustment logic used in randomized trials [67, 68, 96, 47].

Asymptotic linearity and efficiency. Appendix B gives the full influence-pair characterization of this class and the efficient influence pair in the unrestricted two-sample model. For the main paper, the key consequence is that the asymptotic variance of $\sqrt{M}\{\widehat{\psi}(f) - \psi_0\}$ can be written as

$$\mathrm{AVar}\Big\{\sqrt{M}\,(\widehat{\psi}(f) - \psi_0)\Big\} = V_{\mathrm{eff}} + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big],$$

where $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$ is the recentered score and V_{eff} is the semiparametric efficiency bound [10]. Thus, within the AIPW class, efficiency is governed by the mean-squared error of the recentered score $\tilde{f}$ relative to $\mu_0(X)$, and the variance is minimized when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$.

AIPW, PPI, and empirical-efficiency maximization. The usual AIPW estimator corresponds to $f(X) = m(X)$, where $m(X)$ is a prediction score. The original PPI estimator of Angelopoulos et al. [1] corresponds to the scaled score choice $f(X) = m(X)/(1-\rho_n)$, so it is another score choice in the same AIPW class. Relative to AIPW, PPI places the plug-in term entirely on the unlabeled sample and therefore discards labeled-sample covariate information, so it can be less efficient when the original score is already informative.

Because both AIPW and PPI lie in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, it is natural to choose λ data-adaptively rather than fixing it in advance. This yields the estimator called PPI++ by [2]: the AIPW estimator with the empirically rescaled score $\hat{\lambda} m(X)$. Here $\hat{\lambda}$ is chosen by minimizing the empirical variance criterion over $f(X) = \lambda m(X)$, a procedure known as empirical-efficiency maximization [90, 114], and it targets

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{2}$$

The same principle extends beyond simple rescaling: one can minimize the same criterion over richer classes, such as $\{\lambda m(X) + \beta^\top X\}$, monotone score transformations, or an RKHS, and can use it in cross-validation to choose among candidate estimator classes [90]; see Appendix D. For the main paper, the point is simple: AIPW remains safe across the whole class, and our proposal is to improve the score by calibration rather than to change the underlying validity mechanism.

3 Calibrated Prediction-Powered Inference

Section 2 showed that AIPW is safe while the choice of score $f(X)$ determines efficiency. We now introduce calibrated prediction-powered inference, which improves an available prediction score $m(X)$ through post-processing rather than retraining. Using the labeled sample, we learn a transformation of $m(X)$, average the calibrated predictions over the pooled sample, and obtain an estimator that has a simple plug-in form but is exactly AIPW.

3.1 General estimator and exact AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ denote a black-box regression model. A calibration procedure uses the labeled scores $m(X_1), \ldots, m(X_n)$ and outcomes $Y_1, \ldots, Y_n$ to fit a post-hoc map f_n, and then defines the calibrated regression model

$$m_n^\star := f_n \circ m.$$

The resulting predictions on the labeled and unlabeled samples are $m_n^\star(X_i)$ for $i = 1, \ldots, n$ and $m_n^\star(\widetilde{X}_j)$ for $j = 1, \ldots, N$. Intuitively, the map f_n updates the original score $m(X)$ so that its numerical values better align with the outcome.

This post-processing step has two main goals. First, it aims to improve the original prediction score m through a simple transformation, without retraining the underlying model, ideally by improving calibration without sacrificing predictive performance. Second, it is designed so that the resulting plug-in estimator is automatically unbiased, without requiring an explicit residual correction.

Given the post-processed score $m_n^\star$, define the calibrated plug-in estimator by

$$\widehat{\psi}_{\mathrm{cal}} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m_n^\star(X_i) + \sum_{j=1}^{N} m_n^\star(\widetilde{X}_j)\right\}. \tag{3}$$

Equivalently, $\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\}$.

To distinguish how the post-processing map is fit from what property the resulting score satisfies, we introduce two function classes. Let $\mathcal{H}$ be an optimization class of real-valued maps $h : \mathbb{R} \to \mathbb{R}$, and let $\mathcal{F}$ be a linear calibration class of real-valued maps $f : \mathbb{R} \to \mathbb{R}$ that contains the constant functions and the identity map. We fit the post-processing map f_n by empirical risk minimization over $\mathcal{H}$, in the sense that

$$f_n \in \arg\min_{h \in \mathcal{H}} \mathbb{P}_n^L\big[(Y - h(m(X)))^2\big].$$

We then ask what property the resulting post-processed score $m_n^\star := f_n \circ m$ satisfies relative to the calibration class $\mathcal{F}$.

We say that $m_n^\star$ is *empirically calibrated over* $\mathcal{F}$ if, for every $f \in \mathcal{F}$,

$$\mathbb{P}_n^L\big[(Y - m_n^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - f(m_n^\star(X)))^2\big], \tag{4}$$

where $\mathbb{P}_n^L$ denotes the empirical mean over the labeled sample. In words, once $m_n^\star$ has been calibrated, no further transformation of $m_n^\star$ within $\mathcal{F}$ can improve empirical squared-error fit on the labeled sample.

Equivalently, the first-order optimality conditions for (4) are

$$\mathbb{P}_n^L\big[f(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0, \qquad f \in \mathcal{F}. \tag{5}$$

Because $\mathcal{F}$ contains the constant functions and the identity map, these conditions imply, in particular,

$$\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0, \qquad \mathbb{P}_n^L\big[m_n^\star(X)\{Y - m_n^\star(X)\}\big] = 0.$$

The first identity makes the AIPW residual term vanish exactly on the labeled sample.

Thus a calibration method is naturally described by a pair $(\mathcal{H}, \mathcal{F})$: linear and fixed-bin histogram calibration have $\mathcal{H} = \mathcal{F}$, whereas isotonic regression uses a monotone $\mathcal{H}$ but, by Proposition 1, yields a much larger $\mathcal{F}$.

The next result gives the basic structural property of calibrated PPI estimators.

Theorem 1 (Exact AIPW representation under empirical calibration)**.** *Suppose the calibrated score satisfies equation (5), where $\mathcal{F}$ contains all affine maps. Then, for any $\widehat{w} \in \mathcal{F}$,*

$$\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big]. \tag{6}$$

In particular, taking $\widehat{w} \equiv 1$ yields the exact AIPW representation

$$\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m_n^\star(X)\}.$$

If, in addition, $\widehat{w}_n^\star \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\, f(m_n^\star(X))\big] = \rho_n \mathbb{P}_n^L\{f(m_n^\star(X))\} + (1-\rho_n)\mathbb{P}_N^U\{f(m_n^\star(\widetilde{X}))\}, \qquad f \in \mathcal{F}, \tag{7}$$

then

$$\widehat{\psi}_{\mathrm{cal}} = \mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))Y\big]. \tag{8}$$

Theorem 1 turns the pooled plug-in rule into exact AIPW with the calibrated score and, when (7) holds, into an equivalent balancing-weight estimator.

Full versus linear calibration. We focus on two main cases. In *full calibration*, $\mathcal{F}$ is the class of all real-valued transformations, while $\mathcal{H}$ is the fitting class used to learn f_n. This distinction matters because $\mathcal{H}$ serves as a regularizer: without such a restriction, one could overfit the labeled sample and achieve perfect empirical fit with poor population prediction. In practice, $\mathcal{H}$ may consist of piecewise-constant or monotone maps, as in histogram and isotonic calibration. In *linear calibration*, by contrast, $\mathcal{H} = \mathcal{F}$ is the affine class $t \mapsto a + bt$, so both fitting and no-further-calibration are assessed within the same working linear model for Y given $m(X)$.

Why calibration can improve efficiency. Suppose the calibrated score $m_n^\star$ converges to a population limit m_0, where

$$m_0(X) = g_0\{m(X)\}, \qquad g_0 \in \arg\min_{g \in \mathcal{H}} \mathbb{E}_{P_0}\left[\{Y - g(m(X))\}^2\right].$$

Thus, m_0 is the best predictor obtainable from the original score m using post-processing maps in the class $\mathcal{H}$. When $m_n^\star$ is close to m_0, $\widehat{\psi}_{\text{cal}}$ behaves like the AIPW estimator that uses m_0. Proposition 5 therefore implies that this limiting estimator has asymptotic variance no larger than that of the AIPW estimator based on any $\theta \circ m$ with $\theta \in \mathcal{H}$. In particular, if $\mathcal{H}$ contains the identity map, as it does for affine or monotone transformations, then m itself is among the admissible post-processings. It follows that the estimator based on m_0 is asymptotically at least as efficient as the AIPW estimator based on the original score m. Calibration can therefore improve efficiency insofar as m_0 provides a better regression adjustment than m.

3.2 Isotonic-calibrated PPI

We now turn to isotonic regression, the main nonparametric second-stage calibrator. Fitted on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, it is tuning-free, easy to compute, and cannot worsen empirical squared error relative to the original score [127, 71, 6].

The isotonic-calibrated plug-in estimator is

$$\widehat{\psi}_{\text{iso}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m_{n,\text{iso}}^\star(X_i) + \sum_{j=1}^N m_{n,\text{iso}}^\star(\widetilde{X}_j)\right\},$$

where

$$m_{n,\text{iso}}^\star(X) = \hat{f}\{m(X)\}, \qquad \hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i - f(m(X_i))\big\}^2,$$

and $\mathcal{F}_{\text{iso}}$ denotes the class of nondecreasing real-valued functions. The fitted values $m_{n,\text{iso}}^\star(X_i) = \hat{f}(m(X_i))$ are piecewise constant and can be computed by the pooled-adjacent-violators algorithm. This structure implies full empirical calibration, even though the calibration step is restricted to isotonic functions.

Proposition 1 (Isotonic calibration is fully calibrated)**.** *For every function $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\left[h(m_{n,\text{iso}}^\star(X))\{Y - m_{n,\text{iso}}^\star(X)\}\right] = 0. \tag{9}$$

Moreover, it holds that

$$\mathbb{P}_n^L\left[(Y - m_{n,\text{iso}}^\star(X))^2\right] \le \mathbb{P}_n^L\left[(Y - m(X))^2\right].$$

Thus isotonic regression is the main example in which a restricted monotone fitting class $\mathcal{H}$ yields a much larger calibration class $\mathcal{F}$. Consequently, Theorem 1 applies with $\mathcal{F}$ equal to the class of all real-valued functions. Additional implementation details and implementation variants appear in Appendices D and N.

3.3 Linearly calibrated PPI

For smaller labeled samples, one may prefer the lower-dimensional affine case. We therefore focus on linear calibration [66], where the fitting and calibration classes coincide as the affine maps. It

regresses the outcome on the original score and therefore corrects mean bias, scaling, and location shift inside that class.

Let

$$\big(\hat{a}_{\mathrm{lin}}, \hat{b}_{\mathrm{lin}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2, \qquad m^{\star}_{n,\mathrm{lin}}(X) := \hat{a}_{\mathrm{lin}}\, m(X) + \hat{b}_{\mathrm{lin}}. \tag{10}$$

Define the associated linearly calibrated plug-in estimator by

$$\widehat{\psi}_{\mathrm{lin}} := \frac{1}{n+N} \left\{ \sum_{i=1}^{n} m^{\star}_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^{N} m^{\star}_{n,\mathrm{lin}}(\widetilde{X}_j) \right\}. \tag{11}$$

Because the affine class includes the constant function, the least-squares fit automatically enforces empirical mean calibration. As a result, the linearly calibrated plug-in estimator admits an exact AIPW representation.

Proposition 2 (Linear calibration as AIPW)**.** *The normal equations for equation (10) imply*

$$\mathbb{P}_n^L\{Y - m^{\star}_{n,\mathrm{lin}}(X)\} = 0 \qquad \textit{and} \qquad \mathbb{P}_n^L\big[m^{\star}_{n,\mathrm{lin}}(X)\{Y - m^{\star}_{n,\mathrm{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^{\star}_{n,\mathrm{lin}}(X)$*,*

$$\widehat{\psi}_{\mathrm{lin}} = \rho_n \mathbb{P}_n^L\{m^{\star}_{n,\mathrm{lin}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\mathrm{lin}}(\widetilde{X})\} + \mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^{\star}_{n,\mathrm{lin}}(X)\}\big]. \tag{12}$$

The next result connects linear calibration to PPI++, a closely related estimator that chooses a rescaling of the original score by empirical efficiency maximization. In the mean-estimation problem, these two approaches are first-order equivalent, although they need not coincide exactly in finite samples. Intuitively, both are trying to make the same one-dimensional adjustment: they choose the best linear rescaling of $m(X)$. Let $\widehat{\psi}_{++}(\widehat{\lambda}_{++})$ denote the PPI++ estimator of Angelopoulos et al. [2], that is, the AIPW estimator based on the rescaled score $\widehat{\lambda}_{++} m(X)$, where $\widehat{\lambda}_{++}$ is obtained by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ [90, 114]. The key point is that $\widehat{\lambda}_{++}$ estimates λ_0 in (2), which is exactly the population slope in the linear regression of Y on $m(X)$.

Proposition 3 (First-order equivalence of PPI++ and linear calibration)**.** *Assume that* $\mathrm{Var}\{m(X)\} > 0$*,* $\mathbb{E}_{P_0}[Y^2] < \infty$*, and* $\mathbb{E}_{P_0}[m(X)^2] < \infty$*. Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}}\big) \left\{ \mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X) \right\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}} = O_p(n^{-1/2})$*, and, under* $\rho_n \to \rho_0 \in (0,1)$*,*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = O_p\left(\frac{1-\rho_n}{\sqrt{\rho_n}}\, (n+N)^{-1} \right) = O_p\big((n+N)^{-1}\big).$$

In particular,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = o_p\big(M^{-1/2}\big).$$

Linear calibration can thus be viewed as classical prognostic-score adjustment in semisupervised form [41, 84, 65, 96]. Appendix D discusses richer smooth and generalized-linear-model extensions, including Platt scaling and Venn–Abers shrinkage.

3.4 Cross-Fitting and Cross-Calibration

In the main exposition, we assume for simplicity that the initial black-box predictor is fixed. In practice, however, it may be trained on the labeled sample itself, in which case the same labeled observations are used both to fit the predictor and to calibrate its scores. A cross-fitted version replaces m by out-of-fold prediction scores, fits the calibrator on those out-of-fold labeled scores, and then evaluates the calibrated score on the unlabeled sample. This is the natural DML-style implementation when one wishes to reduce overfitting from using the same labeled data both to estimate the initial regression and to calibrate it for inference, so we defer the algorithmic details to the appendix discussion.

4 Theory for isotonic-calibrated PPI

We now study the theoretical basis for *calibeating* via isotonic calibration. Our goal is to understand when post-processing an available score by isotonic regression can improve on the original score and on simpler calibration rules. We establish asymptotic linearity, asymptotic normality, valid Wald inference, and efficiency results, and then show that no further post-processing of the fitted isotonic score yields additional first-order gain. To keep the exposition light, we focus on the non-cross-fitted estimator and treat the initial black-box score m as fixed.

4.1 Asymptotic normality, inference, and efficiency

Throughout this subsection, let f_0 denote a population isotonic calibration map for the fixed score m, defined by

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\big[\{Y - f(m(X))\}^2\big], \qquad m_0 = f_0 \circ m. \tag{13}$$

Thus, m_0 is the population L^2-optimal monotone transformation of the initial score $m(X)$.

Assumption 2 (Regularity conditions for isotonic calibration)**.** Assume:

(i) **Boundedness:** There exists $C_0 < \infty$ such that $|\mathbb{E}_{P_0}[Y \mid m(X)]| \le C_0$ almost surely and $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$.

(ii) **Controlled tails:** The conditional error $Y - \mathbb{E}_{P_0}[Y \mid m(X)]$ is sub-Gaussian or subexponential under P_0.

Under Assumption 2, the isotonic estimator attains the classical rate

$$\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3});$$

see, for example, Chatterjee et al. [14], Yang and Barber [125].

Theorem 2 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold. Then*

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 &= \rho_n(\mathbb{P}^L_n - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ &\quad + (1-\rho_n)(\mathbb{P}^U_N - P_{0,X})\{m_0 - \psi_0\} + O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big).\end{aligned} \tag{14}$$

Thus, under the $\sqrt{M}$ scaling, $\widehat{\psi}_{\text{iso}}$ is asymptotically linear with influence pair $(D^L_{m_0}, D^U_{m_0})$ given by

$$D^L_{m_0}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D^U_{m_0}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

The next corollary records the resulting asymptotic normality and the asymptotic validity of standard Wald confidence intervals. The limiting variance of $\widehat{\psi}_{\text{iso}}$ is

$$\sigma_0^2 := \rho_0 \operatorname{Var}\{D^L_{m_0}(X,Y)\} + (1-\rho_0)\operatorname{Var}\{D^U_{m_0}(\widetilde{X})\}.$$

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 2, assume that $\sigma_0^2 > 0$. Define the empirical variance*

$$\widehat{\sigma}^2 := \rho_n \mathbb{P}^L_n[(\widehat{D}^L)^2] + (1-\rho_n)\mathbb{P}^U_N[(\widehat{D}^U)^2],$$

where

$$\widehat{D}^L_i := m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}, \qquad \widehat{D}^U_j := m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}.$$

Then

$$\frac{\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0)}{\widehat{\sigma}} \xrightarrow{d} N(0,1).$$

Corollary 2 (Efficiency regimes)**.** *Under the conditions of theorem 2, the following hold.*

(i) If $m_0 = \mu_0$ almost surely, then $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

(ii) If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone increasing in $m(X)$, then $m_0(X) = \mathbb{E}_{P_0}[Y \mid m(X)]$ almost surely. Hence $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the reduced two-sample experiment obtained by replacing X with the scalar score $m(X)$.

4.2 No further first-order gain from post-processing

We now ask whether the isotonic-calibrated score can be improved by an additional post-processing step. The key observation is that $\widehat{\psi}_{\text{iso}}$ uses the covariates only through the one-dimensional score $m^{\star}_{n,\text{iso}}(X)$. It is therefore natural to view the estimator as operating in the reduced two-sample experiment

$$\mathcal{E}_{m^{\star}_{n,\text{iso}}} := \Big(\{(m^{\star}_{n,\text{iso}}(X_i), Y_i)\}_{i=1}^{n},\ \{m^{\star}_{n,\text{iso}}(\widetilde{X}_j)\}_{j=1}^{N}\Big),$$

in which the full covariate X is replaced by the scalar score $m^{\star}_{n,\text{iso}}(X)$.

Assumption 3 (Bounded-variation score regression)**.** The map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$ has bounded variation on the support of $m(X)$.

Proposition 4 (No further first-order improvement from post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{\psi}_{\text{eff}} := \widehat{\psi}(\widehat{m}_0), \qquad \widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m^{\star}_{n,\text{iso}}(X)]$$

denote the benchmark estimator in the reduced experiment based on the isotonic score $m^{\star}_{n,\text{iso}}(X)$. Then

$$\|\widehat{m}_0 - m^{\star}_{n,\text{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}),$$

and

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = o_p(M^{-1/2}).$$

In particular, $\widehat{\psi}_{\text{iso}}$ and $\widehat{\psi}_{\text{eff}}$ have the same asymptotic variance.

5 Experiments

5.1 Simulation study

We begin with a controlled binary-outcome simulation designed to isolate score miscalibration. Let $S \sim N(0,1)$, define

$$\mu_0(S) = \{1 + \exp(-5S)\}^{-1},$$

and draw $Y \mid S \sim \text{Bernoulli}\{\mu_0(S)\}$. To induce substantial miscalibration, we apply an affine-biased monotone distortion on the logit scale,

$$m(X) = \text{clip}\left[-0.15 + 0.75\,\sigma\big\{0.8\,z(S) + 0.1\,z(S)^3 - 1\big\},\ 0.01,\ 0.99\right],$$

where $z(S) := \text{logit}\{\mu_0(S)\}$. This preserves ranking information while introducing nonlinear distortion on the probability scale. We compare NAIVE, PPI, AIPW, clipped PPI++, unclipped AIPW-EM, LINEARCAL, MONOSPLINE, ISOCAL, and AUTOCAL; PPI++ uses the official `ppi_py` clipping rule, while AIPW-EM does not. The labeled sample size ranges over $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$, with $N/n \in \{1, 16\}$, and each setting uses 500 Monte Carlo repetitions.

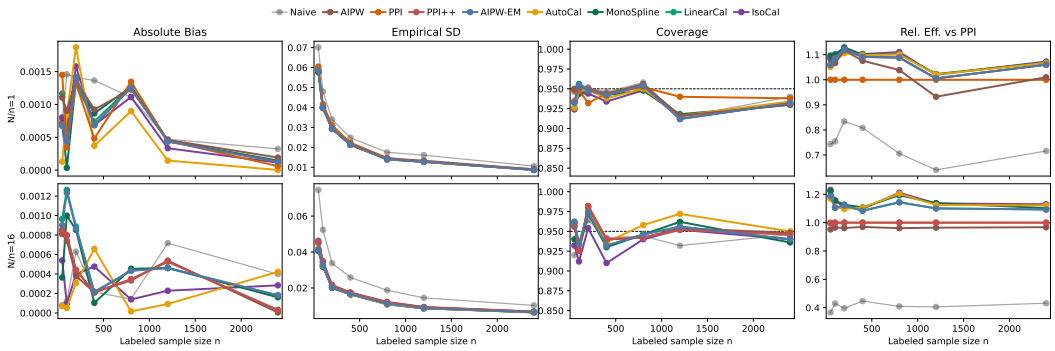

Figure 1: Binary simulation with a miscalibrated score. Columns report bias, Monte Carlo standard deviation, coverage, and relative efficiency versus PPI for $N/n \in \{1, 16\}$.

Figure 1 shows that all score-based AIPW estimators are essentially unbiased, so the differences are mainly in variance. Under miscalibration, especially when $N/n = 16$, calibration improves efficiency. In balanced designs PPI++ and AIPW-EM track LINEARCAL; when the unlabeled sample is large, AIPW-EM stays near LINEARCAL while clipped PPI++ falls back toward PPI. AUTOCAL stays near the best fixed option.

5.2 Empirical Illustration

We evaluate the proposed calibration-based estimators on the mean-estimation benchmarks used in the original experiments of Angelopoulos et al. [1, 2], as provided in the `ppi_py` package. We use the same datasets and labeled-sample-size grids, repeatedly split each fully labeled benchmark into labeled and unlabeled subsets, evaluate all estimators on the same split, treat the full-sample mean as ground truth, and average over 500 random splits. We report `forest`, `galaxies`, and `census_income`.

Baselines are the labeled-only interval, imputation on binary tasks, PPI, AIPW, clipped PPI++, and unclipped AIPW-EM. The main-text figure focuses on PPI, AIPW, PPI++, AIPW-EM, `AutoCal`, and `LinearCal`; Appendix M gives the richer calibration comparison.

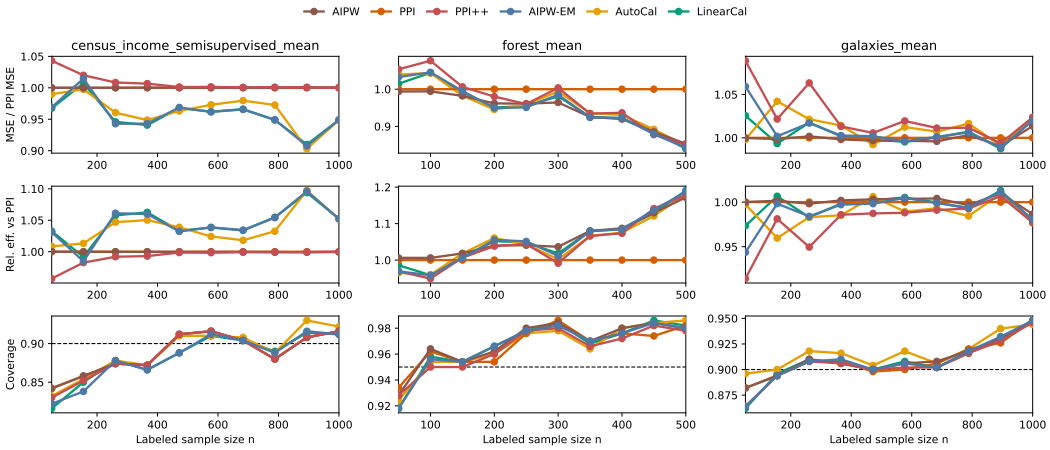

Figure 2: Benchmark summary for PPI, AIPW, clipped PPI++, AIPW-EM, `AutoCal`, and `LinearCal`. Panels report normalized MSE, relative efficiency, and coverage versus PPI.

Figure 2 suggests a simple practical picture. AIPW is the natural estimator based on the original score, and it typically matches or outperforms PPI; when the unlabeled sample is enormous, they nearly coincide, as in `census_income`. Post-hoc calibration is most useful when the score is meaningfully miscalibrated and the labeled sample is large enough to estimate the second-stage calibrator stably, with `LinearCal` strongest on `census_income`. The added AIPW-EM baseline sharpens the comparison with PPI++: once the efficiency-maximization step is left unclipped, it tracks `LinearCal` much more closely, whereas official PPI++ clips the scaling coefficient to $[0, 1]$, motivated by shrinkage estimators, and can therefore fall back toward PPI when the variance-minimizing rescaling would exceed 1. Overall, the practical comparison is that AIPW is the right raw-score baseline, and when miscalibration is substantial the gains come from calibration or unclipped empirical-efficiency maximization rather than from clipped PPI++.

6 Conclusion

We introduced *Calibrated Prediction-Powered Inference* for semisupervised mean estimation. Calibration can improve efficiency without retraining the original model, isotonic calibration can *calibeat* the original score and simpler post-processing rules at first order, and linearly calibrated PPI is first-order equivalent to PPI++ and related to classical prognostic-score regression adjustment.

References

[1] Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023.

[2] Anastasios N Angelopoulos, John C Duchi, and Tijana Zrnic. Ppi++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*, 2023.

[3] Daniele Ballinari and Nora Bearth. Improving the finite sample performance of double/debiased machine learning with propensity score calibration. *arXiv preprint arXiv:2409.04874*, 2024.

[4] Laura B Balzer, Erica Cai, Lucas Godoy Garraza, and Pracheta Amaranath. Adaptive selection of the optimal strategy to improve precision and power in randomized trials. *Biometrics*, 80(1):ujad034, 2024.

[5] Heejung Bang and James M. Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005.

[6] Richard E. Barlow and Hugh D. Brunk. The isotonic regression problem and its dual. *Journal of the American Statistical Association*, 67(337):140–147, 1972.

[7] Antonio Bella, Cèsar Ferri, José Hernández-Orallo, and María José Ramírez-Quintana. Calibration of machine learning models. In *Handbook of Research on Machine Learning Applications and Trends: Algorithms, Methods, and Techniques*, pages 128–146. IGI Global, 2010.

[8] David Benkeser, Marco Carone, MJ Van Der Laan, and Peter B Gilbert. Doubly robust nonparametric inference on the average treatment effect. *Biometrika*, 104(4):863–880, 2017.

[9] Aurélien F. Bibaut and Mark J. van der Laan. Fast rates for empirical risk minimization over càdlàg functions with bounded sectional variation norm. *arXiv preprint arXiv:1907.09244*, 2019.

[10] Peter J. Bickel, Chris A. J. Klaassen, Ya'acov Ritov, and Jon A. Wellner. *Efficient and Adaptive Estimation for Semiparametric Models*. Johns Hopkins University Press, 1993.

[11] David Bruns-Smith, Oliver Dukes, Avi Feller, and Elizabeth L. Ogburn. Augmented balancing weights as linear regression. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, page qkaf019, 2025.

[12] Marco Carone, Iván Díaz, and Mark J van der Laan. Higher-order targeted minimum loss-based estimation. 2014.

[13] Claes M Cassel, Carl E Särndal, and Jan H Wretman. Some results on generalized difference estimation and generalized regression estimation for finite populations. *Biometrika*, 63(3): 615–620, 1976.

[14] Sabyasachi Chatterjee, Adityanand Guntuboyina, and Bodhisattva Sen. On risk bounds in isotonic and other shape restricted regression problems. *The Annals of Statistics*, 43(4):1774–1800, 2015.

[15] Ambarish Chattopadhyay and José R. Zubizarreta. On the implied weights of linear regression for causal inference. *Biometrika*, 110(3):615–629, 2023.

[16] Ambarish Chattopadhyay, Christopher H Hase, and José R Zubizarreta. Balancing vs modeling approaches to weighting in practice. *Statistics in Medicine*, 39(24):3227–3254, 2020.

[17] Qizhao Chen, Vasilis Syrgkanis, and Morgane Austern. Debiased machine learning without sample-splitting for stable estimators. In *Advances in Neural Information Processing Systems*, volume 35, pages 30925–30937, 2022.

[18] Song Xi Chen, Denis HY Leung, and Jing Qin. Improving semiparametric estimation by using surrogate data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 70(4):803–823, 2008.

[19] Yi-Hau Chen. Miscellanea. a robust imputation method for surrogate outcome data. *Biometrika*, 87(3):711–716, 2000.

[20] Yuxi Chen, Edward H Kennedy, and Sivaraman Balakrishnan. On the equivalence between neyman orthogonality and pathwise differentiability. *arXiv preprint arXiv:2603.15817*, 2026.

[21] David Cheng, Ashwin N Ananthakrishnan, and Tianxi Cai. Robust and efficient semi-supervised estimation of average treatment effects with application to electronic health records data. *Biometrics*, 77(2):413–423, 2021.

[22] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

[23] Victor Chernozhukov, Mert Demirer, Esther Duflo, and Ivan Fernandez-Val. Generic machine learning inference on heterogeneous treatment effects in randomized experiments, with an application to immunization in india. Technical report, National Bureau of Economic Research, 2018.

[24] Victor Chernozhukov, Whitney K. Newey, and Rahul Singh. Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027, 2022.

[25] David R Cox. Two further applications of a model for binary regression. *Biometrika*, 45(3/4): 562–565, 1958.

[26] Ilker Demirel, Ahmed Alaa, Anthony Philippakis, and David Sontag. Prediction-powered generalization of causal inferences. *arXiv preprint arXiv:2406.02873*, 2024.

[27] Shachi Deshpande and Volodymyr Kuleshov. Calibrated and conformal propensity scores for causal effect estimation. In *The 40th Conference on Uncertainty in Artificial Intelligence*, 2023.

[28] Iván Díaz. Machine learning in the estimation of causal effects: targeted minimum loss-based estimation and double/debiased machine learning. *Biostatistics*, 21(2):353–358, 2020.

[29] Iván Díaz and Mark J van der Laan. Doubly robust inference for targeted minimum loss–based estimation in randomized trials with missing outcome data. *Statistics in medicine*, 36 (24):3807–3819, 2017.

[30] Peng Ding and Fan Li. Causal inference: A missing data perspective, 2018. URL https://arxiv.org/abs/1712.06170.

[31] Dean P. Foster and Sergiu Hart. "Calibeating": Beating Forecasters at Their Own Game. *Theoretical Economics*, 18(4):1441–1474, 2023. doi: 10.3982/TE5330.

[32] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2):243–268, 2007.

[33] Ellen Graham, Marco Carone, and Andrea Rotnitzky. Towards a unified theory for semiparametric data fusion with individual-level data. *arXiv preprint arXiv:2409.09973*, 2024.

[34] Susan Gruber and Mark J Van Der Laan. Targeted maximum likelihood estimation: A gentle introduction. *U.C. Berkeley Division of Biostatistics Working Paper Series*, 2009.

[35] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.

[36] Chirag Gupta and Aaditya Ramdas. Distribution-free calibration guarantees for histogram binning without sample splitting. In *International conference on machine learning*, pages 3942–3952. PMLR, 2021.

[37] Chirag Gupta and Aaditya Ramdas. Online platt scaling with calibeating. In *International Conference on Machine Learning*, pages 12182–12204. PMLR, 2023.

[38] Rom Gutman, Ehud Karavani, and Yishai Shimoni. Propensity score models are better when post-calibrated. *arXiv preprint arXiv:2211.01221*, 2022.

[39] Jens Hainmueller. Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46, 2012.

[40] Peisong Han. A note on improving the efficiency of inverse probability weighted estimator using the augmentation term. *Statistics & Probability Letters*, 82(12):2221–2228, 2012.

[41] Ben B Hansen. The prognostic analogue of the propensity score. *Biometrika*, 95(2):481–488, 2008.

[42] Rafail Z Hasminskii and Ildar A Ibragimov. On the nonparametric estimation of functionals. In *Proceedings of the Second Prague Symposium on Asymptotic Statistics*, volume 473, pages 474–482. North-Holland Amsterdam, 1979.

[43] Nima S. Hejazi and Mark J. van der Laan. Revisiting the propensity score's central role: Towards bridging balance and efficiency in the era of causal machine learning. *Observational Studies*, 9(1):23–34, 2023.

[44] Nima S Hejazi, Mark J van der Laan, Holly E Janes, Peter B Gilbert, and David C Benkeser. Efficient nonparametric inference on the effects of stochastic interventions under two-phase sampling, with applications to vaccine efficacy trials. *Biometrics*, 77(4):1241–1253, 2021.

[45] Katherine Hoffman. An illustrated guide to tmle, part i: Introduction and motivation, December 2020. URL https://www.khstats.com/blog/tmle/tutorial.

[46] Emilie Højbjerre-Frandsen and Alejandro Schuler. "within-trial" prognostic score adjustment is targeted maximum likelihood estimation. *Pharmaceutical Statistics*, 25(2):e70080, 2026.

[47] Emilie Højbjerre-Frandsen, Mark J van der Laan, and Alejandro Schuler. Powering rcts for marginal effects with glms using prognostic score adjustment. *arXiv preprint arXiv:2503.22284*, 2025.

[48] Chuan Hong, Katherine P. Liao, and Tianxi Cai. Semi-supervised validation of multiple surrogate outcomes with application to electronic medical records phenotyping. *Biometrics*, 75(1):78–89, 2019. doi: 10.1111/biom.12971.

[49] Kosuke Imai and Marc Ratkovic. Covariate balancing propensity score. *Journal of the Royal Statistical Society Series B*, 76(1):243–263, 2014.

[50] Wenlong Ji, Lihua Lei, and Tijana Zrnic. Predictions as surrogates: Revisiting surrogate outcomes in the age of ai. *arXiv preprint arXiv:2501.09731*, 2025.

[51] Man Jin. Prognostic covariate adjustment in non-ideal conditions: Limitations and doubly robust alternatives. *Statistics in Biopharmaceutical Research*, (just-accepted):1–33, 2026.

[52] Christopher Jung, Changhwa Lee, Mallesh Pai, Aaron Roth, and Rakesh Vohra. Moment multicalibration for uncertainty estimation. In *Conference on Learning Theory*, pages 2634–2678. PMLR, 2021.

[53] Nathan Kallus and Xiaojie Mao. On the role of surrogates in the efficient estimation of treatment effects with limited outcome data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 87(2):480–509, 2025.

[54] Edward H Kennedy. Semiparametric doubly robust targeted double machine learning: a review. *Handbook of statistical methods for precision medicine*, pages 207–236, 2024.

[55] Chris AJ Klaassen. Consistent estimation of the influence function of locally asymptotically linear estimators. *The Annals of Statistics*, 15(4):1548–1562, 1987.

[56] Nicolas S Lambert. Elicitation and evaluation of statistical forecasts. *Preprint*, 3(5):18, 2011.

[57] Donghwan Lee, Xinmeng Huang, Hamed Hassani, and Edgar Dobriban. T-cal: An optimal test for the calibration of predictive models. *Journal of Machine Learning Research*, 24(335): 1–72, 2023.

[58] Se Yoon Lee and Jae-Kwang Kim. Mec: Machine-learning-assisted generalized entropy calibration for semi-supervised mean estimation. *arXiv preprint arXiv:2604.05446*, 2026.

[59] Samuel D. Lendle, Bruce Fireman, and Mark J. van der Laan. Balancing score adjusted targeted minimum loss-based estimation. *Journal of Causal Inference*, 3(2):139–155, 2015.

[60] Yan Leng and Drew Dimmery. Calibration of heterogeneous treatment effects in random experiments. *Available at SSRN 3875850*, 2021.

[61] Boris Yakovlevich Levit. On efficiency of a class of non-parametric estimates. *Teoriya Veroyatnostei i ee Primeneniya*, 20(4):738–754, 1975.

[62] Sijia Li and Alex Luedtke. Efficient estimation under data fusion. *Biometrika*, 110(4):1041–1054, 2023.

[63] Sijia Li, Peter B Gilbert, Rui Duan, and Alex Luedtke. Data fusion using weakly aligned sources. *Journal of the American Statistical Association*, 120(552):2569–2579, 2025.

[64] Sarah Lichtenstein, Baruch Fischhoff, and Lawrence D Phillips. Calibration of probabilities: The state of the art. In *Decision Making and Change in Human Affairs: Proceedings of the Fifth Research Conference on Subjective Probability, Utility, and Decision Making, Darmstadt, 1–4 September, 1975*, pages 275–324. Springer, 1977.

[65] Winston Lin. Agnostic notes on regression adjustments to experimental data: Reexamining freedman's critique. *The Annals of Applied Statistics*, pages 295–318, 2013.

[66] Jacob A Mincer and Victor Zarnowitz. The evaluation of economic forecasts. In *Economic forecasts and expectations: Analysis of forecasting behavior and performance*, pages 3–46. NBER, 1969.

[67] Kelly L Moore and Mark J van der Laan. Covariate adjustment in randomized trials with binary outcomes: targeted maximum likelihood estimation. *Statistics in medicine*, 28(1): 39–64, 2009.

[68] Kelly L Moore and Mark J van der Laan. Increasing power in randomized trials with right censored outcomes through covariate adjustment. *Journal of biopharmaceutical statistics*, 19 (6):1099–1131, 2009.

[69] Kelly L Moore, Romain Neugebauer, Thamban Valappil, and Mark J van der Laan. Robust extraction of covariate information to improve estimation efficiency in randomized trials. *Statistics in medicine*, 30(19):2389–2408, 2011.

[70] Reagan Mozer. Ppi is the difference estimator: Recognizing the survey sampling roots of prediction-powered inference. *arXiv preprint arXiv:2603.19160*, 2026.

[71] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 625–632, 2005.

[72] Georgy Noarov and Aaron Roth. The statistical scope of multicalibration. In *International Conference on Machine Learning*, pages 26283–26310. PMLR, 2023.

[73] Margaret Sullivan Pepe. Inference using surrogate outcome data and a validation sample. *Biometrika*, 79(2):355–365, 1992.

[74] Margaret Sullivan Pepe, Marie Reilly, and Thomas R Fleming. Auxiliary outcome data and the mean score method. *Journal of Statistical Planning and Inference*, 42(1-2):137–160, 1994.

[75] J Pfanzagl and W Wefelmeyer. Contributions to a general asymptotic statistical theory. *Statistics & Risk Modeling*, 3(3-4):379–388, 1985.

[76] John Platt et al. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in large margin classifiers*, 10(3):61–74, 1999.

[77] Kristin E Porter, Susan Gruber, Mark J Van Der Laan, and Jasjeet S Sekhon. The relative performance of targeted maximum likelihood estimators. *The international journal of biostatistics*, 7(1):31, 2011.

[78] Pierre-Emmanuel Poulet, Maylis Tran, Sophie Tezenas du Montcel, Bruno Dubois, Stanley Durrleman, Bruno Jedynak, and Alzheimer's Disease Neuroimaging Initiative. Prediction-powered inference for clinical trials: application to linear covariate adjustment. *BMC Medical Research Methodology*, 25(1):204, 2025.

[79] Sky Qiu, Susan Gruber, Pamela A Shaw, Brian D Williamson, and Mark J van der Laan. Efficient targeted maximum likelihood estimators for two-phase design problems. *arXiv preprint arXiv:2602.24131*, 2026.

[80] James M Robins and Andrea Rotnitzky. Semiparametric efficiency in multivariate regression models with missing data. *Journal of the American Statistical Association*, 90(429):122–129, 1995.

[81] James M Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American statistical Association*, 89(427):846–866, 1994.

[82] James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Analysis of semiparametric regression models for repeated outcomes in the presence of missing data. *Journal of the American Statistical Association*, 90(429):106–121, 1995.

[83] Sherri Rose and Mark J van der Laan. A targeted maximum likelihood estimator for two-stage designs. *The international journal of biostatistics*, 7(1):17, 2011.

[84] Michael Rosenblum and Mark J van der Laan. Using regression models to analyze randomized trials: Asymptotically valid hypothesis tests despite incorrectly specified models. *Biometrics*, 65(3):937–945, 2009.

[85] Rachael K Ross, Lina M Montoya, Dana E Goin, Iván Díaz, and Audrey Renson. Constructing targeted minimum loss/maximum likelihood estimators: a simple illustration to build intuition. *American journal of epidemiology*, page kwaf261, 2025.

[86] Aaron Roth. Uncertain: Modern topics in uncertainty estimation. *Unpublished Lecture Notes*, 11(30-31):4, 2022.

[87] Andrea Rotnitzky, Quanhong Lei, Mariela Sued, and James M Robins. Improved double-robust estimation in missing data and causal inference models. *Biometrika*, 99(2):439–456, 2012.

[88] Andrea Rotnitzky, Ezequiel Smucler, and James M Robins. Characterization of parameters with a mixed bias property. *Biometrika*, 108(1):231–238, 2021.

[89] Andrea Rotnitzky, Ezequiel Smucler, and James M Robins. A note on the relation between one-step, outcome regression and ipw-type estimators of parameters with the mixed bias property. *arXiv preprint arXiv:2509.22452*, 2025.

[90] Daniel B Rubin and Mark J van der Laan. Empirical efficiency maximization: improved locally efficient covariate adjustment in randomized experiments and survival analysis. *The International Journal of Biostatistics*, 4(1), 2008.

[91] Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976.

[92] Donald B Rubin. Bayesian inference for causal effects: The role of randomization. *The Annals of statistics*, pages 34–58, 1978.

[93] Carl-Erik Särndal, Bengt Swensson, and Jan Wretman. *Model assisted survey sampling*. Springer Science & Business Media, 2003.

[94] Daniel O Scharfstein, Andrea Rotnitzky, and James M Robins. Adjusting for nonignorable drop-out using semiparametric nonresponse models. *Journal of the American Statistical Association*, 94(448):1096–1120, 1999.

[95] Anton Schick. On asymptotically efficient estimation in semiparametric models. *The Annals of Statistics*, pages 1139–1151, 1986.

[96] Alejandro Schuler, David Walsh, Diana Hall, Jon Walsh, Charles Fisher, Critical Path for Alzheimer's Disease, Alzheimer's Disease Neuroimaging Initiative, and Alzheimer's Disease Cooperative Study. Increasing the efficiency of randomized trial estimates via linear adjustment for a prognostic score. *The International Journal of Biostatistics*, 18(2):329–356, 2022.

[97] Andrew J Scott and D Holt. The effect of two-stage sampling on ordinary least squares methods. *Journal of the American Statistical Association*, 77(380):848–854, 1982.

[98] Shaun R Seaman and Stijn Vansteelandt. Introduction to double robust methods for incomplete data. *Statistical science: a review journal of the Institute of Mathematical Statistics*, 33 (2):184, 2018.

[99] Yilin Song, Dan M Kluger, Harsh Parikh, and Tian Gu. Demystifying prediction powered inference. *arXiv preprint arXiv:2601.20819*, 2026.

[100] Zhou Tang and Ted Westling. Consistency of the bootstrap for asymptotically linear estimators based on machine learning. *arXiv preprint arXiv:2404.03064*, 2024.

[101] Anastasios A Tsiatis. *Semiparametric theory and missing data*. Springer, 2006.

[102] Lars van der Laan and Ahmed Alaa. Generalized venn and venn-abers calibration with applications in conformal prediction. *arXiv preprint arXiv:2502.05676*, 2025.

[103] Lars van der Laan and Ahmed M Alaa. Self-calibrating conformal prediction. *Advances in Neural Information Processing Systems*, 37:107138–107170, 2024.

[104] Lars van der Laan and Nathan Kallus. Bellman calibration for v-learning in offline reinforcement learning. *arXiv preprint arXiv:2512.23694*, 2025.

[105] Lars van der Laan, Marco Carone, Alex Luedtke, and Mark van der Laan. Adaptive debiased machine learning using data-driven model selection techniques. *arXiv preprint arXiv:2307.12544*, 2023.

[106] Lars van der Laan, Ernesto Ulloa-Pérez, Marco Carone, and Alex Luedtke. Causal isotonic calibration for heterogeneous treatment effects. In *International Conference on Machine Learning*, pages 34831–34854, 2023.

[107] Lars van der Laan, Ziming Lin, Marco Carone, and Alex Luedtke. Stabilized inverse probability weighting via isotonic calibration. *arXiv preprint arXiv:2411.06342*, 2024.

[108] Lars van der Laan, Alex Luedtke, and Marco Carone. Doubly robust inference via calibration. *arXiv preprint arXiv:2411.02771*, 2024.

[109] Lars van der Laan, Aurelien Bibaut, Nathan Kallus, and Alex Luedtke. Automatic debiased machine learning for smooth functionals of nonparametric m-estimands. *arXiv preprint arXiv:2501.11868*, 2025.

[110] Lars van der Laan, David Hubbard, Allen Tran, Nathan Kallus, and Aurélien Bibaut. Semiparametric double reinforcement learning with applications to long-term causal inference. *arXiv preprint arXiv:2501.06926*, 2025.

[111] Mark van der Laan. Cv-tmle and double machine learning, December 2019. URL https://vanderlaan-lab.org/2019/12/24/cv-tmle-and-double-machine-learning/.

[112] Mark van der Laan, Zeyi Wang, and Lars van der Laan. Higher order targeted maximum likelihood estimation. *arXiv preprint arXiv:2101.06290*, 2021.

[113] Mark J Van der Laan. Targeted estimation of nuisance parameters to obtain valid statistical inference. *The international journal of biostatistics*, 10(1):29–57, 2014.

[114] Mark J. van der Laan and James M. Robins. *Unified Methods for Censored Longitudinal Data and Causality*. Springer, 2003.

[115] Mark J. van der Laan and Sherri Rose. *Targeted Learning: Causal Inference for Observational and Experimental Data*. Springer, 2011.

[116] Mark J Van Der Laan and Daniel Rubin. Targeted maximum likelihood learning. *U.C. Berkeley Division of Biostatistics Working Paper Series*, 2006.

[117] Aad W Van der Vaart. *Asymptotic statistics*, volume 3. Cambridge university press, 2000.

[118] Adriaan W. van der Vaart and Jon A. Wellner. *Weak Convergence and Empirical Processes*. Springer, 1996.

[119] Vladimir Vovk and Ivan Petej. Venn-abers predictors. *arXiv preprint arXiv:1211.0025*, 2012.

[120] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic learning in a random world*. Springer, 2005.

[121] Volodimir G Vovk. Universal forecasting algorithms. *Information and Computation*, 96(2): 245–277, 1992.

[122] Cheng Wang. Calibration in deep learning: A survey of the state-of-the-art. *arXiv preprint arXiv:2308.01222*, 2023.

[123] J Whitehouse, C Jung, V Syrgkanis, B Wilder, and ZS Wu. Orthogonal causal calibration. *arXiv preprint arXiv:2406.01933*, 2024.

[124] Zichun Xu, Daniela Witten, and Ali Shojaie. A unified framework for semiparametrically efficient semi-supervised learning. *arXiv preprint arXiv:2502.17741*, 2025.

[125] Fan Yang and Rina Foygel Barber. Contraction and uniform convergence of isotonic regression. *arXiv preprint arXiv:1706.01852*, 2018.

[126] Bianca Zadrozny and Charles Elkan. Obtaining calibrated probability estimates from decision trees and naive bayesian classifiers. In *Proceedings of the Eighteenth International Conference on Machine Learning*, pages 609–616, 2001.

[127] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 694–699, 2002.

[128] Yichi Zhang, Tianrun Cai, Sheng Yu, Kelly Cho, Chuan Hong, Jiehuan Sun, Jie Huang, Yuk-Lam Ho, Ashwin N. Ananthakrishnan, Zongqi Xia, Stanley Y. Shaw, Vivian Gainer, Victor Castro, Nicholas Link, Jacqueline Honerlaw, Sicong Huang, David Gagnon, Elizabeth W. Karlson, Robert M. Plenge, Peter Szolovits, Guergana Savova, Susanne Churchill, Christopher O'Donnell, Shawn N. Murphy, J. Michael Gaziano, Isaac Kohane, Tianxi Cai, and Katherine P. Liao. High-throughput phenotyping with electronic medical record data using a common semi-supervised approach (phecap). *Nature Protocols*, 14(12):3426–3444, 2019. doi: 10.1038/s41596-019-0227-6.

[129] Wenjing Zheng and Mark J Van Der Laan. Asymptotic theory for cross-validated targeted maximum likelihood estimation. *U.C. Berkeley Division of Biostatistics Working Paper Series*, 2010.

[130] Tijana Zrnic and Emmanuel J. Candès. Cross-prediction-powered inference. *Proceedings of the National Academy of Sciences*, 121(15):e2322083121, 2024.

[131] José R Zubizarreta. Stable weights that balance covariates for estimation with incomplete outcome data. *Journal of the American Statistical Association*, 110(511):910–922, 2015.

A Extended related work

Although PPI has been presented as a recent development, the underlying problem and many of its estimators are special cases of a broader literature on semiparametric debiasing and regression adjustment, including work on missing data, causal inference, and debiased machine learning.

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin [91] formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes. Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures [61, 75, 42, 55, 10]. In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations [81, 82, 80, 114, 101], often with doubly robust structure [5]. As a result, the efficiency theory for missing-data models under missingness or coarsening at random is now well understood, and the class of regular asymptotically linear estimators in these models can be characterized through their influence functions [81, 82, 80, 114, 101].

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine learning. Flexible learning methods within estimating equations and AIPW-style procedures already appear in van der Laan and Robins [114]. Targeted minimum loss estimation (TMLE), or targeted learning, updates an initial nuisance estimator in a parameter-targeted way so that the resulting plug-in estimator is debiased; in the present setting, the main nuisance is the outcome regression [116, 115, 34, 45, 85]. These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine-learning estimators [95, 55, 129, 115, 22]. Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators [22, 24], and can be understood within the same semiparametric framework [111, 28, 20]. For sufficiently stable estimators, cross-fitting can itself be relaxed [17]. Closely related work studies balancing-weight estimators [39, 49, 131, 59, 16, 43], which often admit equivalent regression-adjustment or augmented representations [15, 11, 89]. In completely missing-at-random settings, including semisupervised learning and randomized trials, these approaches allow flexible black-box nuisance estimation while still supporting valid inference under appropriate conditions [67, 68, 84, 47, 46].

Semisupervised inference. Related work studies semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes or surrogate measurements are available more broadly [73, 74, 19, 18, 21, 50, 53]. Two-sample semisupervised inference is a special case of two-stage sampling designs [97, 83, 44, 79] and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed [62, 63, 33, 124]. More recent papers have used the term *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample [1, 2, 130, 124, 99, 78, 58]. These papers also helped popularize such methods for a broader machine-learning audience and provided accessible software. Closely related questions have long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical or auxiliary data sources are used to improve precision in smaller trials [41, 84, 67–69, 96, 4, 47]. Methodologically, however, PPI estimators are best understood as instances or variants of existing semiparametric procedures. For example, PPI++ is an AIPW estimator [81, 114] obtained by empirical efficiency maximization [90] over a scaling parameter, and cross-PPI [130] is simply AIPW with cross-fitting. MEC [58] instead uses generalized entropy weights to balance the prediction score, which is closer to the balancing-weight representation of our linearly calibrated estimator. Our linearly calibrated estimator emphasizes the corresponding regression-adjustment view, implemented by ordinary linear regression of Y on the score. More fundamentally, the ability to combine black-box prediction models with valid inference is a consequence of the double robustness of AIPW when the missingness probability is known, and is not specific to semisupervised mean estimation [5, 98, 88].

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is reinterpreted as a treatment or sampling indicator, so that the relevant potential outcomes are viewed as missing [92, 114, 30, 70]. This equivalence allows tools from covariate adjustment, causal in-

ference, and model-assisted estimation to be transferred directly to the semisupervised setting; see, for example, Mozer [70] for results relating prediction-powered inference to model-assisted survey-sampling estimators. The two-sample formulation we study is also equivalent to a one-sample i.i.d. model [62]. In particular, our methods apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized trial with binary treatment, where treated units form the labeled sample and control units contribute covariate information only; see Appendix E.

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling [76, 25, 37], linear calibration [66, 23, 60], isotonic regression [127, 71], and histogram binning [126, 36] improve predictive reliability without retraining the underlying model. The idea has deeper roots in the forecasting literature [66, 64, 121, 120, 32, 56]; see also the historical discussion in Lee et al. [57]. More recent work studies calibration under broader losses and prediction tasks [52, 72, 106, 123, 102], as well as applications to conformal prediction [103, 102], importance-weight stabilization [38, 27, 3, 107], treatment effect estimation [23, 106, 123], and value prediction in offline reinforcement learning [104]. Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas [36], while van der Laan et al. [106], van der Laan and Alaa [102] establish asymptotic distribution-free guarantees for isotonic calibration. Most closely related to our work is recent research connecting calibration to debiased estimation of statistical functionals [108, 105, 110]. In particular, van der Laan et al. [108] studies calibrated debiased machine learning for doubly robust inference, which in our setting reduces to calibrating the outcome regression and exploiting the resulting plug-in representation, while van der Laan et al. [110] studies calibrated plug-in regression estimators in reinforcement learning. More broadly, van der Laan et al. [105] shows how plug-in calibration can support regression and treatment effect estimation, including superefficient inference through dimension reduction.

B Semiparametric background and equivalence material

B.1 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model [81]. It is helpful to view AIPW not as a single estimator, but as a family of estimators indexed by a score function $f : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such f, define

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1-\rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{15}$$

This estimator combines a plug-in term that imputes Y from $f(X)$ with a residual correction from the labeled sample. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for $\psi_0 = \mathbb{E}_{P_0}[Y]$. In particular, $f(X) = 0$ yields the empirical mean estimator $\mathbb{P}_n^L\{Y\}$. Thus, the choice of f affects efficiency, through the estimator's variance, but not validity, in the sense that the resulting confidence intervals remain valid. This makes AIPW a "safe" estimator [124]. This property can be viewed as a particularly simple instance of double robustness[1] [5]. The same basic principle underlies the use of flexible machine learning for covariate adjustment in randomized trials while retaining valid inference [67, 68, 96, 47].

Asymptotic linearity and efficiency. To characterize the leading large-sample error in the two-sample model, we decompose the estimation error into one contribution from the labeled sample and one from the unlabeled sample. We say that $\widehat{\psi}$ is asymptotically linear with influence pair (D^L, D^U) if

$$\widehat{\psi} - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

where $\mathbb{E}_{P_0}[D^L(X,Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D^U(\widetilde{X})] = 0$. By the central limit theorem and the independence of the two samples,

$$\sqrt{M}\,(\widehat{\psi} - \psi_0) \rightsquigarrow N\Big(0,\ \rho_0 \mathrm{Var}_{P_0}\{D^L(X,Y)\} + (1-\rho_0)\mathrm{Var}_{P_{0,X}}\{D^U(\widetilde{X})\}\Big).$$

[1] In more general missing-at-random settings with unknown, covariate-dependent missingness probabilities, the analogous AIPW estimator remains doubly robust asymptotically, but it no longer enjoys the same finite-sample unbiasedness as in the present known-ρ_n setting [98].

The influence pair therefore determines the estimator's asymptotic variance and provides a convenient way to compare efficiency across different choices of f. The next result uses this representation to characterize the AIPW class, identify the efficient influence pair in the unrestricted two-sample model, and thereby obtain the semiparametrically efficient variance bound [10].

Proposition 5 (Influence pair for the AIPW class)**.** *Under Assumption 1, for any fixed $f : \mathcal{X} \to \mathbb{R}$ with finite second moment, the estimator $\widehat{\psi}(f)$ is asymptotically linear with influence pair (D_f^L, D_f^U), where*

$$D_f^L(X, Y) = \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\}, \tag{16}$$

$$D_f^U(\widetilde{X}) = \tilde{f}(\widetilde{X}) - \psi_0, \tag{17}$$

with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$ denoting the recentered version of $f(X)$. The resulting asymptotic variance of $\sqrt{M}\,(\widehat{\psi}(f) - \psi_0)$ is

$$\mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

This variance is minimized over the AIPW class when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$. Therefore, the efficient influence pair for $\psi_0 = \mathbb{E}_{P_0}[Y]$ in the unrestricted two-sample model is $(D_{\mu_0}^L, D_{\mu_0}^U)$, and the corresponding asymptotic variance is the efficiency bound[2]*.*

A key insight of Proposition 5 is that the limiting variance of the AIPW estimator depends on the score function f only through the mean squared error of its recentered version $\tilde{f}$, namely $\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big]$. Thus, within the AIPW class, efficiency is maximized by choosing f so that $\tilde{f}$ approximates $\mu_0(X)$ as well as possible, with equality when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$. This shows that full efficiency generally requires using the full covariate information X, or at least a summary rich enough to recover $\mu_0(X)$ [80]. In practice, this motivates taking f to be an estimated prediction score $m(X)$, learned using flexible black-box methods.

AIPW, PPI, and empirical-efficiency maximization. We now record several familiar choices of f within the class (15). The most common choice is $f(X) = m(X)$, where $m(X)$ is a prediction score, yielding the usual augmented inverse-probability weighted (AIPW) estimator [81]:

$$\widehat{\psi}_{\mathrm{AIPW}} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Equivalently,

$$\widehat{\psi}_{\mathrm{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^{n} m(X_i) + \sum_{j=1}^{N} m(\widetilde{X}_j) + \frac{1}{\rho_n}\sum_{i=1}^{n}\{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the inverse probability of observing the outcome.

Another estimator in this class was considered by Angelopoulos et al. [1] under the name PPI,[3] and we adopt that terminology here. It corresponds to the score choice

$$f(X) = \frac{m(X)}{1-\rho_n}.$$

Substituting this choice into (15) gives

$$\widehat{\psi}_{\mathrm{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

Relative to $\widehat{\psi}_{\mathrm{AIPW}}$, $\widehat{\psi}_{\mathrm{PPI}}$ uses only the unlabeled sample in the plug-in term and rescales the score by the factor $(1-\rho_n)^{-1} > 1$. By Proposition 5, this helps only if the rescaled score is closer to $\mu_0(X)$ in mean squared error. Thus, PPI is inefficient when $m = \mu_0$, and is typically less efficient

[2] Appendix B.2 shows that every regular estimator is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators and the semiparametric efficiency bound in this model.

[3] The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. [13] and Särndal et al. [93], respectively; see Mozer [70] for a review.

when m is already reasonably accurate. When ρ_n is small and m is sufficiently inaccurate, however, neither estimator is necessarily more efficient than the other [40, 87].

Because both $\widehat{\psi}_{\mathrm{AIPW}}$ and $\widehat{\psi}_{\mathrm{PPI}}$ use scores in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, it is natural to choose the scaling factor data-adaptively rather than fixing it in advance. One way to do so is to minimize the estimator's empirical influence-function variance over this class:

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L \left[\left\{ f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X)) \right\}^2 \right] + (1-\rho_n)\mathbb{P}_N^U \left[\left\{ f(\widetilde{X}) - \widehat{\psi}(f) \right\}^2 \right].$$

That is, one chooses λ to minimize $\sigma_n^2(f)$ over $f(X) = \lambda m(X)$, a procedure known as empirical-efficiency maximization [90, 114]. The resulting estimator, called PPI++ by Angelopoulos et al. [2], is simply the AIPW estimator with an empirically rescaled prediction rule $\hat{\lambda} m(X)$, where $\hat{\lambda}$ estimates the population-optimal scaling coefficient

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{18}$$

The same principle extends beyond simple rescaling. For example, one can minimize $\sigma_n^2(f)$ over richer classes such as $\{\lambda m(X) + \beta_{\mathrm{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\mathrm{lin}} \in \mathbb{R}^d\}$, over monotone transformations of the score, or over a reproducing kernel Hilbert space (Appendix D). More generally, the same criterion can be used in cross-validation to choose among candidate estimator classes [90].

B.2 Additional background on semiparametric statistics

The next result shows that the efficient choice in Proposition 5 is not merely optimal within the AIPW class. In fact, every regular estimator[4] in the unrestricted two-sample model is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators at first order in this model.

Proposition 6 (Characterization of regular estimators)**.** *Under Assumption 1, a mean-zero pair (D^L, D^U) is the influence pair of a regular estimator of ψ_0 in the unrestricted two-sample model if and only if there exists a function $f_0 : \mathcal{X} \to \mathbb{R}$ with finite second moment, potentially depending on P_0, such that*

$$(D^L, D^U) = (D_{f_0}^L, D_{f_0}^U),$$

where D_f^L and D_f^U are given in (16)–(17). Equivalently, for every regular estimator $\widehat{\psi}^\dagger$, there exists such an f_0 for which

$$\widehat{\psi}^\dagger - \widehat{\psi}(f_0) = o_p(M^{-1/2}).$$

This characterization shows that the AIPW class is the relevant first-order benchmark in this model. Appendix B.2 illustrates that several estimators not always written in AIPW form—including regression adjustment, balancing-weight adjustment, and TMLE—can nonetheless be viewed as finite-sample or asymptotic equivalents of AIPW estimators [89, 116], while sometimes offering better finite-sample performance [77, 51, 109].

Remark 1 (Plug-in approaches to bias correction and TMLE)**. Regression plug-in estimators.** A particularly simple member of this first-order class is obtained through plug-in regression adjustment [94, 116, 84, 89]. Let

$$m_n(X) = \hat{a}_{\mathrm{lin}}\, m(X) + \hat{b}_{\mathrm{lin}}, \qquad (\hat{a}_{\mathrm{lin}}, \hat{b}_{\mathrm{lin}}) = \arg\min_{a,b} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2.$$

The corresponding plug-in estimator is

$$\rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

For any $\lambda \in \mathbb{R}$, this estimator also admits the AIPW representation

$$\rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\} + \lambda \mathbb{P}_n^L\big[Y - m_n(X)\big],$$

[4] Informally, a regular estimator is one whose large-sample behavior remains stable under small changes in the data-generating distribution [117].

because the augmentation term $\mathbb{P}_n^L[Y - m_n(X)]$ is zero by the least-squares normal equations. Thus, when m_n converges to the best linear predictor of Y given $m(X)$, this plug-in estimator is typically at least as efficient as $\widehat{\psi}_{\mathrm{AIPW}}$ and $\widehat{\psi}_{\mathrm{PPI}}$, since it corresponds to an influence function with smaller or equal variance.

Balancing-weight estimators. The same estimator can also be written in terms of balancing weights. Let $w_n(X) = \hat{a}_{\mathrm{bal}} + \hat{b}_{\mathrm{bal}}\, m(X)$ be affine weights that balance $(1, m(X))$, in the sense that

$$\mathbb{P}_n^L\{w_n(X)\} = 1 \quad \text{and} \quad \mathbb{P}_n^L\big[m(X)w_n(X)\big] = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\}.$$

Then the corresponding balancing-weight estimator satisfies

$$\mathbb{P}_n^L\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\big[w_n(X)\{Y - m(X)\}\big].$$

In fact, the plug-in and balancing-weight estimators are algebraically equivalent [15, 11]:

$$\mathbb{P}_n^L\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

TMLE. More broadly, such debiased plug-in estimators can be viewed as instances of TMLE [116, 115, 46], a general framework for constructing plug-in estimators that are debiased and, in many cases, efficient by updating nuisance components to remove first-order bias. In the present setting, either the weights or the regression model are updated so that the corresponding augmentation term vanishes. In weighting-based TMLE, the update balances on the prediction model $m(X)$ (e.g., Hejazi and van der Laan 43); one may either include the known weight $1/\rho_n$ as an offset or balance on the intercept as well. In regression-based TMLE, the prediction model $m(X)$ may be included either as a covariate or as an offset (e.g., Lendle et al. 59), and the intercept adjustment accounts for the known inverse missingness-probability weight $1/\rho_n$. The specific targeting step is ultimately a practical choice, provided it solves the relevant estimating equation and removes the augmentation term; different choices may also yield additional benefits, such as further bias reduction or doubly robust inference [113, 12, 29, 8, 112, 108].

C Constructing confidence intervals via the bootstrap

Confidence intervals may be constructed using the standard Wald approximation from Corollary 1, namely

$$\widehat{\psi}_{\mathrm{iso}} \pm z_{1-\alpha/2}\, \frac{\hat{\sigma}}{\sqrt{n+N}}.$$

As an alternative, one may use the nonparametric bootstrap and refit the isotonic calibration step within each bootstrap replicate, as proposed and theoretically studied by van der Laan et al. [108]; see also Tang and Westling [100]. This avoids direct estimation of the influence-function variance and automatically propagates uncertainty from estimation of the calibration map.

For each bootstrap replicate $b = 1, \ldots, B$, we sample with replacement from the labeled observations $\{(X_i, Y_i)\}_{i=1}^n$ and, independently, from the unlabeled observations $\{\widetilde{X}_j\}_{j=1}^N$, obtaining bootstrap samples

$$\{(X_i^{*(b)}, Y_i^{*(b)})\}_{i=1}^n, \qquad \{\widetilde{X}_j^{*(b)}\}_{j=1}^N.$$

We then refit the isotonic calibrator on the bootstrap labeled sample:

$$\hat{f}^{*(b)} \in \arg\min_{f \in \mathcal{F}_{\mathrm{iso}}} \sum_{i=1}^n \big\{Y_i^{*(b)} - f(m(X_i^{*(b)}))\big\}^2,$$

and define

$$m_{\mathrm{iso}}^{*(b)}(X) := \hat{f}^{*(b)}\{m(X)\}.$$

The corresponding bootstrap replicate of the estimator is

$$\widehat{\psi}_{\mathrm{iso}}^{*(b)} := \frac{1}{n+N}\left\{\sum_{i=1}^n m_{\mathrm{iso}}^{*(b)}(X_i^{*(b)}) + \sum_{j=1}^N m_{\mathrm{iso}}^{*(b)}(\widetilde{X}_j^{*(b)})\right\}.$$

After computing B bootstrap replicates, one may form a percentile interval using the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of

$$\{\widehat{\psi}_{\mathrm{iso}}^{*(b)}\}_{b=1}^{B}.$$

Alternatively, one may use the bootstrap standard error

$$\widehat{\sigma}_{\mathrm{boot}}^2 := \frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\psi}_{\mathrm{iso}}^{*(b)} - \bar{\psi}_{\mathrm{iso}}^{*}\right)^2, \qquad \bar{\psi}_{\mathrm{iso}}^{*} := \frac{1}{B}\sum_{b=1}^{B}\widehat{\psi}_{\mathrm{iso}}^{*(b)},$$

and report the normal-approximation interval

$$\widehat{\psi}_{\mathrm{iso}} \pm z_{1-\alpha/2}\,\widehat{\sigma}_{\mathrm{boot}}.$$

The key implementation detail is that the isotonic regression step must be refit within each bootstrap replicate rather than held fixed. Otherwise, the resulting interval does not reflect uncertainty from estimation of the calibration map.

D Additional remarks

Remark 2 (Implementation)**.** Isotonic regression is widely available in standard software. A standard algorithm is the pooled-adjacent-violators algorithm (PAVA) [6]. Although it is generally fast, it may become slow at ultra-large sample sizes. A scalable alternative is to use regression tree software with monotonicity constraints, such as XGBoost or LightGBM (see Appendix N). An advantage of tree-based software is that the fit can be regularized by constraining or tuning the number of leaves, the maximum tree depth, or the minimum number of observations per leaf.

Remark 3 (Platt scaling adjustment)**.** The linear-calibration idea extends beyond squared-error loss. One can instead use logistic loss, Poisson loss, or, more generally, losses from canonical-link generalized linear models [84]. In particular, when $Y \in \{0,1\}$ and $m(X) \in (0,1)$ is an initial predictor, one may apply Platt scaling [25, 76] by fitting a logistic regression of Y on the logit of $m(X)$:

$$\left(\hat{a}_{\mathrm{platt}}, \hat{b}_{\mathrm{platt}}\right) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n}\Big[-Y_i \log\sigma\big(a\,\mathrm{logit}(m(X_i))+b\big) - (1-Y_i)\log\Big(1-\sigma\big(a\,\mathrm{logit}(m(X_i))+b\big)\Big)\Big], \tag{19}$$

with calibrated predictor

$$m_{n,\mathrm{platt}}^{\star}(X) := \sigma\big(\hat{a}_{\mathrm{platt}}\,\mathrm{logit}(m(X)) + \hat{b}_{\mathrm{platt}}\big),$$

where $\sigma(t) := (1+e^{-t})^{-1}$. Platt scaling may be preferred over linear regression calibration when predicted probabilities are systematically over- or under-confident, in which case the calibration curve is often approximately sigmoidal. More generally, outcomes can be rescaled to $[0,1]$ before applying this procedure. In online settings, calibration properties of Platt scaling are developed in [37].

Remark 4 (Empirical efficiency maximization as a learning objective)**.** Following Rubin and van der Laan [90], the empirical variance criterion from Section B.1,

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\Big\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\Big\}^2\right] + (1-\rho_n)\mathbb{P}_N^U\left[\Big\{f(\widetilde{X}) - \widehat{\psi}(f)\Big\}^2\right],$$

is not tied to the one-dimensional scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$. For any candidate class $\mathcal{G} \subset L^2(P_{0,X})$, one may instead define

$$\hat{f} \in \arg\min_{f\in\mathcal{G}} \sigma_n^2(f), \qquad \widehat{\psi}_{\mathrm{eem}} := \widehat{\psi}(\hat{f}),$$

so empirical efficiency maximization learns the augmentation $f(X)$ directly.

At the population level, let

$$V(f) := \rho_0 \mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\}$$

denote the asymptotic variance of $\widehat{\psi}(f)$. Using equations (16) and (17),

$$V(f) = \mathrm{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}\{Y - f(X)\}.$$

Equivalently, with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$,

$$V(f) = \mathrm{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathbb{E}\big[(Y - \tilde{f}(X))^2\big].$$

Thus the efficiency-optimal choice is the $L^2(P_{0,X})$-projection of μ_0 onto the normalized class $\{\tilde{f} : f \in \mathcal{G}\}$. If $\mathcal{G}$ is closed under additive constants, then minimizing $V(f)$ is equivalent to ordinary least-squares regression of Y on X over $\mathcal{G}$. In particular, rather than optimizing the empirical efficiency criterion $\sigma_n^2(f)$ directly, one can equivalently optimize the usual empirical mean-squared-error objective on the labeled sample.

This makes richer efficiency-maximizing classes immediate. For example, if $\mathcal{G} = \{b + h : h \in \mathcal{H},\ b \in \mathbb{R}\}$ for an RKHS $\mathcal{H}$ with kernel K, then a penalized version of empirical efficiency maximization is

$$(\hat{b}_\lambda, \hat{h}_\lambda) \in \arg\min_{b\in\mathbb{R},\ h\in\mathcal{H}} \left\{\mathbb{P}_n^L\big[(Y - b - h(X))^2\big] + \lambda\|h\|_{\mathcal{H}}^2\right\},$$

which is just kernel ridge regression on the labeled sample, up to the irrelevant additive constant in the variance criterion. By the representer theorem,

$$\hat{h}_\lambda(\cdot) = \sum_{i=1}^n \hat{\alpha}_i K(X_i, \cdot),$$

and, after centering the kernel or including the intercept explicitly,

$$\hat{\alpha} = (K_n + n\lambda I_n)^{-1}(Y - \hat{b}_\lambda \mathbf{1}_n).$$

Plugging $\hat{f}_\lambda = \hat{b}_\lambda + \hat{h}_\lambda$ into $\widehat{\psi}(\hat{f}_\lambda)$ yields the corresponding efficiency-maximized AIPW estimator.

For tuning and class selection, it is preferable to evaluate the same criterion out of fold. One may split the labeled sample into folds, fit f^{-k} on the training part of fold k, compute the held-out criterion $\sigma_{n,k}^2(f^{-k})$, and average over folds. The selected class or tuning parameter can then be used in a cross-fitted or cross-averaged AIPW estimator. This gives a direct cross-validated empirical-efficiency objective for choosing among classes such as linear, spline, isotonic, or RKHS-based regressors while preserving the usual sample-splitting protection against in-sample overfitting.

Remark 5 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification [119]; see [103, 102] for generalized versions beyond the binary setting. Assume $Y \in [0,1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0,1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\big[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\big].$$

Following [119], one may convert this interval to a point prediction by shrinking its midpoint toward the (unadjusted) AIPW estimator:

$$m^*_{n,\mathrm{VA}}(x) := m^*_{n,\mathrm{mid}}(x) + \big\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\big\}\big\{\widehat{\psi}_{AIPW} - m^*_{n,\mathrm{mid}}(x)\big\},$$

where

$$m^*_{n,\mathrm{mid}}(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

Since the shrinkage magnitude is proportional to the width of the Venn–Abers interval, this construction can be more stable than plain isotonic calibration in small samples.

Remark 6 (Relation between linear calibration, PPI, and prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through flexible covariate adjustment in small-sample settings [41, 84, 65, 96, 4, 47].

Proposition 3 shows that the PPI++ estimator admits the same first-order interpretation. Likewise, the PPI estimator $\widehat{\psi}_{\mathrm{PPI}}$ is algebraically equivalent to the unlabeled-sample plug-in estimator $\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$ based on the intercept-only adjustment

$$\hat{a}_{\mathrm{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\mathrm{const}}(X) := \hat{a}_{\mathrm{const}} + m(X), \tag{20}$$

which calibrates only the mean. By contrast, the standard AIPW estimator $\widehat{\psi}_{\mathrm{AIPW}}$ is equivalent to the pooled-sample mean $\rho_n \mathbb{P}_n^L\{m^\star_{n,\mathrm{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$.

Remark 7 (Practical takeaway)**.** Isotonic calibration can improve efficiency when the original score is informative but miscalibrated. The resulting estimator behaves as if $m(X)$ were replaced by its best monotone calibration $m_0(X)$. Because the isotonic class contains the identity map and all affine maps with positive slope, the isotonic-calibrated estimator has limiting variance no larger than that of any estimator that is first-order equivalent to AIPW with score $\theta\{m(X)\}$ for such a transformation θ. This includes AIPW and PPI based on $m(X)$, as well as PPI++, and linear calibration whenever the corresponding scaling or slope coefficient is nonnegative. Thus isotonic calibration is asymptotically at least as efficient, at first order, as these methods, and can be strictly more efficient when the best monotone calibration is genuinely nonlinear. More generally, if $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone, then no regular estimator based only on $m(X)$ can improve on the isotonic-calibrated estimator at first order.

Remark 8 (When full efficiency is attained)**.** Full efficiency requires that the calibrated target recover all regression information relevant to ψ_0. This holds if $m_0 = \mu_0$ almost surely. More generally, full efficiency fails when the one-dimensional score $m(X)$ omits predictive information in X that cannot be recovered by monotone calibration.

E Prediction-Powered Causal Inference via Calibration (i.e., Calibrated DML)

This appendix records the direct translation of the two-sample missing-outcome setup to randomized experiments. The central point is that the semisupervised problem studied in the main text and two-arm causal inference in a randomized trial are the same statistical problem once the unobserved potential outcome in one arm is interpreted as the missing outcome. We then illustrate how our calibrated plug-in estimators apply in this setting, where they arise as special cases of calibrated DML [108].

We use the standard potential-outcomes notation. Each unit has two potential outcomes, $Y(1)$ under treatment and $Y(0)$ under control, and the observed outcome satisfies

$$Y = AY(1) + (1-A)Y(0),$$

where $A \in \{0,1\}$ is the treatment assignment. The mean potential outcomes are

$$\mu_1 := \mathbb{E}\{Y(1)\}, \qquad \mu_0 := \mathbb{E}\{Y(0)\},$$

and the average treatment effect is $\tau := \mu_1 - \mu_0$. In a randomized trial, identification follows from the usual causal conditions: consistency and no interference, exchangeability induced by randomization,

$$A \perp (X, Y(1), Y(0)),$$

and positivity, so that each arm has positive assignment probability.

Now partition the observed data into the two arm-specific samples

$$\mathcal{D}_{n,1} := \{(X_i, Y_i) : A_i = 1\}, \qquad \mathcal{D}_{n,0} := \{(\widetilde{X}_j, \widetilde{Y}_j) : A_j = 0\},$$

with sample sizes $n_1 := |\mathcal{D}_{n,1}|$, $n_0 := |\mathcal{D}_{n,0}|$, and $n = n_1 + n_0$. Let $\mathbb{P}_{n,a}$ denote the empirical mean over arm a, and let $\hat{\pi}_a := n_a/n$. To estimate μ_1, one treats $\mathcal{D}_{n,1}$ as the labeled sample for $Y(1)$ and uses the covariates in $\mathcal{D}_{n,0}$ as the unlabeled sample. To estimate μ_0, one reverses the roles of the two arms. Thus each mean potential outcome is a two-sample missing-outcome estimand of exactly the form analyzed in the main text.

For $a \in \{0,1\}$, let $m_a(X)$ be an arm-specific prediction score fit using only observations in arm a. The standard arm-specific AIPW estimator is

$$\hat{\mu}_a^{\mathrm{AIPW}} := \mathbb{P}_n\{m_a(X)\} + \mathbb{P}_n\left[\frac{\mathbf{1}(A=a)}{\hat{\pi}_a}\{Y - m_a(X)\}\right].$$

For $a \in \{0,1\}$, let $m_{n,a}^*(X)$ denote the calibrated version of the score $m_a(X)$. The calibrated estimator of μ_a is then

$$\hat{\mu}_a^{\mathrm{cal}} := \mathbb{P}_n\{m_{n,a}^*(X)\} = \hat{\pi}_a\,\mathbb{P}_{n,a}\{m_{n,a}^*(X)\} + (1-\hat{\pi}_a)\,\mathbb{P}_{n,1-a}\{m_{n,a}^*(X)\}. \tag{21}$$

This is exactly the pooled plug-in estimator from the two-sample formulation, with one arm contributing outcomes and the other contributing only covariates. The same estimator also has the exact AIPW form

$$\hat{\mu}_a^{\mathrm{cal}} = \mathbb{P}_n\{m_{n,a}^*(X)\} + \mathbb{P}_n\left[\frac{\mathbf{1}(A=a)}{\hat{\pi}_a}\{Y - m_{n,a}^*(X)\}\right]. \tag{22}$$

Because

$$\mathbb{P}_n\left[\frac{\mathbf{1}(A=a)}{\hat{\pi}_a}\{Y - m_{n,a}^*(X)\}\right] = \mathbb{P}_{n,a}\{Y - m_{n,a}^*(X)\},$$

the residual correction vanishes exactly whenever the calibrator includes the constant score and therefore enforces mean calibration within arm a. This is the randomized-trial analogue of the exact augmented representation in theorem 1.

The average treatment effect is then estimated by

$$\hat{\tau}^{\mathrm{cal}} := \hat{\mu}_1^{\mathrm{cal}} - \hat{\mu}_0^{\mathrm{cal}}. \tag{23}$$

Equivalently,

$$\hat{\tau}^{\mathrm{cal}} = \mathbb{P}_n\{\tilde{m}_1(X) - \tilde{m}_0(X)\} + \mathbb{P}_n\left[\frac{A}{\hat{\pi}_1}\{Y - \tilde{m}_1(X)\} - \frac{1-A}{\hat{\pi}_0}\{Y - \tilde{m}_0(X)\}\right].$$

Its influence function is the difference of the two arm-specific AIPW influences,

$$D_\tau(O) = \left\{\tilde{m}_1(X) - \mu_1 + \frac{A}{\pi_1}\big(Y - \tilde{m}_1(X)\big)\right\} - \left\{\tilde{m}_0(X) - \mu_0 + \frac{1-A}{\pi_0}\big(Y - \tilde{m}_0(X)\big)\right\}.$$

Accordingly, Wald inference for the ATE may be based on the empirical variance of the estimated influence values divided by n, or on the nonparametric bootstrap refitting the calibration step within each bootstrap sample.

The practical implementation is therefore immediate. One splits the trial into $\mathcal{D}_{n,1}$ and $\mathcal{D}_{n,0}$, treats one arm as labeled and the other as unlabeled when estimating a given mean potential outcome, fits and calibrates an arm-specific score, averages the calibrated predictions over the pooled covariate sample, and finally subtracts the two arm-specific estimates to obtain the ATE. In this way, the semisupervised mean estimators in the main text become direct estimators of mean potential outcomes and treatment effects in randomized experiments.

F General Moment Equations Under Missing at Random

We briefly review the general AIPW class for moment-equation targets under missing at random, following Robins et al. [81, 82], Robins and Rotnitzky [80], van der Laan and Robins [114]. Let

$$O = (X, R, RY),$$

where $R \in \{0,1\}$ indicates whether Y is observed, and suppose

$$R \perp Y \mid X, \qquad \pi_0(X) := \mathbb{P}(R=1 \mid X) > 0 \text{ a.s.}$$

Let the target $\theta_0 \in \Theta \subseteq \mathbb{R}^d$ be identified by

$$\mathbb{E}_{P_0}[\varphi(Y, \theta_0)] = 0,$$

where $\varphi : \mathcal{Y} \times \Theta \to \mathbb{R}^d$ is a known identifying function. Define

$$A_0 := \partial_\theta \mathbb{E}_{P_0}[\varphi(Y,\theta)]\big|_{\theta=\theta_0},$$

and assume A_0 is nonsingular.

Theorem 3 (AIPW class for general moment equations under MAR)**.** *For any square-integrable function $m : \mathcal{X} \times \Theta \to \mathbb{R}^d$, define*

$$\psi(O; \theta, m, \pi_0) := m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}. \tag{24}$$

Then

$$\mathbb{E}[\psi(O; \theta_0, m, \pi_0)] = 0.$$

Hence the equation

$$\mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) = 0$$

defines an unbiased AIPW estimating equation for θ_0.

Moreover, let $\hat{\theta}$ be a regular $\sqrt{n}$-consistent solution of this equation, and assume that $\theta \mapsto \mathbb{E}[\psi(O; \theta, m, \pi_0)]$ is differentiable at θ_0 with derivative A_0, and that

$$(\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big] = o_p\big(n^{-1/2} + \|\hat{\theta} - \theta_0\|\big).$$

Then $\hat{\theta}$ has influence function

$$D_m(O) = -A_0^{-1}\psi(O; \theta_0, m, \pi_0) = -A_0^{-1}\left[m(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m(X, \theta_0)\}\right]. \tag{25}$$

Thus the class of regular Z-estimators arising from (24) is indexed by the choice of augmentation function m.

The efficient choice in the full MAR model is

$$m_0(X, \theta_0) := \mathbb{E}[\varphi(Y, \theta_0) \mid X], \tag{26}$$

which yields the efficient influence function

$$D_{\text{eff}}(O) = -A_0^{-1}\left[m_0(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\}\right]. \tag{27}$$

Proof of theorem 3. For any $\theta \in \Theta$,

$$\begin{aligned}
\mathbb{E}\{\psi(O; \theta, m, \pi_0)\} &= \mathbb{E}\left[m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\left[m(X, \theta) + \frac{\mathbb{E}(R \mid X, Y)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\left[m(X, \theta) + \frac{\mathbb{E}(R \mid X)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}\big[\varphi(Y, \theta)\big],
\end{aligned}$$

where the third line uses $R \perp Y \mid X$. Evaluating at $\theta = \theta_0$ gives

$$\mathbb{E}\{\psi(O; \theta_0, m, \pi_0)\} = \mathbb{E}\big[\varphi(Y, \theta_0)\big] = 0.$$

Let

$$\Psi_m(\theta) := \mathbb{E}\{\psi(O; \theta, m, \pi_0)\}.$$

The previous calculation shows that $\Psi_m(\theta) = \mathbb{E}[\varphi(Y, \theta)]$ for every θ, so

$$\partial_\theta \Psi_m(\theta)\big|_{\theta=\theta_0} = A_0.$$

Because $\hat{\theta}$ solves the empirical estimating equation,

$$0 = \mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) = (\mathbb{P}_n - P_0)\psi(O; \theta_0, m, \pi_0) + \{\Psi_m(\hat{\theta}) - \Psi_m(\theta_0)\} + (\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big].$$

By differentiability of Ψ_m at θ_0,

$$\Psi_m(\hat{\theta}) - \Psi_m(\theta_0) = A_0(\hat{\theta} - \theta_0) + o(\|\hat{\theta} - \theta_0\|).$$

Since $\hat{\theta}$ is $\sqrt{n}$-consistent, the previous display and the stated remainder condition imply

$$A_0(\hat{\theta} - \theta_0) = -(\mathbb{P}_n - P_0)\psi(O; \theta_0, m, \pi_0) + o_p(n^{-1/2}).$$

Multiplying by $-A_0^{-1}$ gives

$$\hat{\theta} - \theta_0 = (\mathbb{P}_n - P_0)D_m(O) + o_p(n^{-1/2}),$$

with D_m given in equation (25). This proves the influence-function formula.

Now define

$$m_0(X, \theta_0) := \mathbb{E}\{\varphi(Y, \theta_0) \mid X\}, \qquad \delta_m(X) := m(X, \theta_0) - m_0(X, \theta_0).$$

Then

$$D_m(O) = D_{\text{eff}}(O) - A_0^{-1}\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X).$$

Fix any $u \in \mathbb{R}^d$. Because $\delta_m(X)$ is measurable with respect to X,

$$\mathbb{E}\left[\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X) \,\middle|\, X\right] = 0.$$

Also, using $R \perp Y \mid X$ and $\mathbb{E}[\varphi(Y, \theta_0) - m_0(X, \theta_0) \mid X] = 0$,

$$\mathbb{E}\left[\frac{R}{\pi_0(X)}\left(1 - \frac{R}{\pi_0(X)}\right)\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\} \,\middle|\, X\right] = 0.$$

Therefore the cross-term vanishes:

$$\mathbb{E}\left[\{u^\top D_{\text{eff}}(O)\}\left(1 - \frac{R}{\pi_0(X)}\right)\{u^\top A_0^{-1}\delta_m(X)\}\right] = 0.$$

Moreover,

$$\mathbb{E}\left[\left(1 - \frac{R}{\pi_0(X)}\right)^2 \,\middle|\, X\right] = \frac{1 - \pi_0(X)}{\pi_0(X)}.$$

Expanding the variance of $u^\top D_m(O)$ therefore yields

$$\text{Var}\{u^\top D_m(O)\} = \text{Var}\{u^\top D_{\text{eff}}(O)\} + \mathbb{E}\left[\frac{1 - \pi_0(X)}{\pi_0(X)}\{u^\top A_0^{-1}\delta_m(X)\}^2\right].$$

The second term is nonnegative and equals zero if and only if $\delta_m(X) = 0$ almost surely. Hence $m_0(X, \theta_0)$ minimizes the asymptotic variance of every linear contrast, so D_{eff} is the efficient influence function within this class. The missing-data efficiency results of Robins et al. [81, 82], Robins and Rotnitzky [80], van der Laan and Robins [114] identify this same influence function as the semiparametric efficient influence function in the full MAR model. □

Corollary 3 (Two-sample/PPI form)**.** *Suppose the labeled and unlabeled samples arise from the semisupervised model with constant labeling probability ρ_0, and let $S = f(X)$ be a chosen score. Replacing X by S gives the reduced AIPW class*

$$\psi(O; \theta, m, \rho_0) = m(S, \theta) + \frac{R}{\rho_0}\{\varphi(Y, \theta) - m(S, \theta)\}.$$

In the two-sample notation used in the main text, the empirical estimating equation based on the known labeled fraction $\rho_n = n/(n + N)$ can be written as

$$0 = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\} + \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\}.$$

The efficient choice within the reduced score model is

$$m_0(S, \theta_0) = \mathbb{E}[\varphi(Y, \theta_0) \mid S].$$

Proof. Because the semisupervised design has constant labeling probability, R is independent of (X, Y), hence also independent of Y given $S = f(X)$, and

$$\mathbb{P}(R = 1 \mid S) = \rho_0.$$

Applying theorem 3 with covariate S and observation probability $\pi_0(S) \equiv \rho_0$ gives the displayed reduced AIPW class and the efficient choice

$$m_0(S, \theta_0) = \mathbb{E}[\varphi(Y, \theta_0) \mid S].$$

At the sample level, the two-sample implementation replaces the population labeling probability ρ_0 by the known design fraction $\rho_n = n/(n+N)$. To rewrite the empirical equation in the two-sample notation, note that

$$\mathbb{P}_M\{m(S, \theta)\} = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\},$$

where $\mathbb{P}_M$ denotes the empirical mean over the pooled sample. Also,

$$\mathbb{P}_M\left[\frac{R}{\rho_n}\{\varphi(Y, \theta) - m(S, \theta)\}\right] = \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\},$$

because $R = 1$ on labeled observations and $R = 0$ on unlabeled observations. Adding the two displays yields exactly

$$0 = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\} + \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\}.$$

□

G Proof of the representation theorem

Proof of theorem 1. Because equation (5) holds for every $h \in \mathcal{F}$, any empirical balancing weight $\widehat{w}(m_n^\star(X)) \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Adding this zero term to the plug-in estimator equation (3) yields equation (6).

For the second claim, since the identity map belongs to $\mathcal{F}$, taking $f(t) = t$ in equation (7) gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} = \widehat{\psi}_{\text{cal}}.$$

Since $\widehat{w}_n^* \in \mathcal{F}$, equation (5) also gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Therefore,

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))Y\big] = \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] + \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = \widehat{\psi}_{\text{cal}},$$

which is equation (8). □

Lemma 1 (Blockwise residual orthogonality)**.** *Let $B_1, \ldots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $m_{n,\text{iso}}^\star(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \ldots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(m_{n,\text{iso}}^\star(X_i))$ is finite for all i,

$$\sum_{i=1}^{n} h(m_{n,\text{iso}}^\star(X_i))\{Y_i - m_{n,\text{iso}}^\star(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(m_{n,\text{iso}}^\star(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^{n} h(m_{n,\text{iso}}^\star(X_i))\{Y_i - m_{n,\text{iso}}^\star(X_i)\} = \sum_{j=1}^{J} h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 1. The orthogonality claim equation (9) is exactly the conclusion of Lemma 1. For the risk comparison, note that the identity map belongs to $\mathcal{F}_{\text{iso}}$. Since $\hat{f}$ minimizes the empirical squared loss over $\mathcal{F}_{\text{iso}}$,

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\text{iso}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - \hat{f}_{\text{id}}(m(X)))^2\big] = \mathbb{P}_n^L\big[(Y - m(X))^2\big],$$

where $\hat{f}_{\text{id}}(t) = t$. □

Proof of proposition 2. The normal equations for equation (10) are

$$\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} = 0 \qquad \text{and} \qquad \mathbb{P}_n^L\big[m(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Because $m^\star_{n,\text{lin}}(X) = \hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}$, these imply

$$\mathbb{P}_n^L\big[m^\star_{n,\text{lin}}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^\star_{n,\text{lin}}(X)$,

$$\mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = a\,\mathbb{P}_n^L\{Y - m^\star_{n,\text{lin}}(X)\} + b\,\mathbb{P}_n^L\big[m^\star_{n,\text{lin}}(X)\{Y - m^\star_{n,\text{lin}}(X)\}\big] = 0.$$

Adding this zero term to equation (11) yields equation (12). □

H Proofs of first-order equivalence between PPI++ and linear calibration

Recall that

$$(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2,$$

and define the corresponding linear-calibration estimator by

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\}.$$

Also let $\widehat{\psi}_{++}$ denote the PPI++ estimator obtained by minimizing the empirical influence-function variance over the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, and let $\hat{\lambda}_{++}$ be the resulting selected coefficient. Define

$$\Delta_n := \mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X),$$

and

$$a_0 := \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}}.$$

Proof of Proposition 3. The normal equations for the linear regression fit are

$$\mathbb{P}_n^L\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\} = 0$$

and

$$\mathbb{P}_n^L\big[m(X)\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\}\big] = 0.$$

The first equation gives

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X).$$

Substituting this into the second equation yields

$$\hat{a}_{\text{lin}} = \frac{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))(Y - \mathbb{P}_n^L Y)\big]}{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big]}.$$

Hence $\hat{a}_{\text{lin}}$ is the ordinary least-squares slope, and under finite second moments,

$$\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2}).$$

Using

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X),$$

we obtain

$$\begin{aligned}
\widehat{\psi}_{\text{lin}} &= \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\} \\
&= \hat{b}_{\text{lin}} + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y - \hat{a}_{\text{lin}}\mathbb{P}_n^L m(X) + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\Delta_n.
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) &= \rho_n \mathbb{P}_n^L\{\hat{\lambda}_{++} m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{\lambda}_{++} m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - \hat{\lambda}_{++} m(X)\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\Delta_n.
\end{aligned}$$

Therefore,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n.$$

Under the empirical PPI++ variance criterion over the class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, the selected coefficient satisfies

$$\hat{\lambda}_{++} = \frac{\mathbb{P}_n^L\big[(Y - \mathbb{P}_n^L Y)(m(X) - \mathbb{P}_n^L m(X))\big]}{(1-\rho_n)\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big] + \rho_n \mathbb{P}_N^U\big[(m(\widetilde{X}) - \mathbb{P}_N^U m(\widetilde{X}))^2\big]}.$$

Its population limit is

$$\lambda_0 = \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}} = a_0,$$

so standard M-estimation arguments yield

$$\hat{\lambda}_{++} = a_0 + O_p(n^{-1/2}).$$

Combining this with $\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2})$ gives

$$\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2}).$$

Next, since $\mathbb{P}_n^L m(X)$ and $\mathbb{P}_N^U m(\widetilde{X})$ are sample means under the same marginal law of X,

$$\Delta_n = O_p(n^{-1/2} + N^{-1/2}).$$

Under the standing assumption $\rho_n = n/(n+N) \to \rho_0 \in (0,1)$, we may write

$$n^{-1/2} = \rho_n^{-1/2}(n+N)^{-1/2}, \qquad N^{-1/2} = (1-\rho_n)^{-1/2}(n+N)^{-1/2},$$

and hence

$$\Delta_n = O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big).$$

Therefore,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} &= (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n \\
&= (1-\rho_n)\, O_p(n^{-1/2})\, O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big) \\
&= O_p\Big((1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1}\Big).
\end{aligned}$$

Since

$$(1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\} = \frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}},$$

It follows that

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\Big(\Big\{\frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}}\Big\}(n+N)^{-1}\Big).$$

In particular, this implies the simpler bound

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\left(\frac{1-\rho_n}{\rho_n}\,(n+N)^{-1}\right).$$

Therefore, under $\rho_n \to \rho_0 \in (0,1)$,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\big((n+N)^{-1}\big) = o_p\big((n+N)^{-1/2}\big).$$

This proves the claim. □

Proposition 7 (PPI and AIPW as intercept-only calibration)**.** *Let*

$$\hat{a}_{\text{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^{\star}_{n,\text{const}}(X) := \hat{a}_{\text{const}} + m(X).$$

Then

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Consequently, the PPI estimator

$$\widehat{\psi}_{\text{PPI}} := \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{PPI}} = \mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\},$$

while the standard AIPW estimator

$$\widehat{\psi}_{\text{AIPW}} := \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{AIPW}} = \rho_n \mathbb{P}_n^L\{m^{\star}_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\}.$$

Hence PPI is the unlabeled-only plug-in estimator based on the mean-calibrated score $m^{\star}_{n,\text{const}}$, whereas AIPW is the corresponding pooled plug-in estimator.

Proof. The objective in the definition of $\hat{a}_{\text{const}}$ is quadratic in a, and its first-order condition is

$$0 = \mathbb{P}_n^L\{Y - a - m(X)\}.$$

Therefore

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Using $m^{\star}_{n,\text{const}}(\widetilde{X}) = m(\widetilde{X}) + \hat{a}_{\text{const}}$, we obtain

$$\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} = \widehat{\psi}_{\text{PPI}}.$$

Likewise,

$$\begin{aligned}
\rho_n \mathbb{P}_n^L\{m^{\star}_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\} &= \rho_n \mathbb{P}_n^L\{m(X) + \hat{a}_{\text{const}}\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X}) + \hat{a}_{\text{const}}\} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} \\
&= \widehat{\psi}_{\text{AIPW}}.
\end{aligned}$$

This proves the claim. □

I Proof of Proposition 5

Proof of proposition 5. Let

$$c := \psi_0 - \mathbb{E}_{P_{0,X}}\{f(X)\}.$$

Because c is constant,

$$\widehat{\psi}(f + c) = \widehat{\psi}(f) + \rho_n c + (1 - \rho_n)c - c = \widehat{\psi}(f).$$

Therefore $\widehat{\psi}(f) = \widehat{\psi}(\tilde{f})$, where

$$\tilde{f}(X) = f(X) + c = f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0.$$

By construction,

$$\mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0.$$

Using $\mathbb{E}_{P_0}[Y] = \psi_0$ and $\mathbb{E}_{P_0}\{\tilde{f}(X)\} = \mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0$, we obtain

$$\begin{aligned}
\widehat{\psi}(f) - \psi_0 &= \widehat{\psi}(\tilde{f}) - \psi_0 \\
&= \rho_n(\mathbb{P}_n^L - P_0)\{\tilde{f}(X)\} + (1 - \rho_n)(\mathbb{P}_N^U - P_{0,X})\{\tilde{f}(\widetilde{X})\} + (\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} \\
&= \rho_n(\mathbb{P}_n^L - P_0)\Big\{\tilde{f}(X) - \psi_0 + \rho_n^{-1}\big(Y - \tilde{f}(X)\big)\Big\} \\
&\quad + (1 - \rho_n)(\mathbb{P}_N^U - P_{0,X})\big\{\tilde{f}(\widetilde{X}) - \psi_0\big\}.
\end{aligned}$$

Define

$$D_{f,n}^L(X, Y) := \tilde{f}(X) - \psi_0 + \rho_n^{-1}\{Y - \tilde{f}(X)\}, \qquad D_f^U(\widetilde{X}) := \tilde{f}(\widetilde{X}) - \psi_0.$$

Then

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D_{f,n}^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D_f^U(\widetilde{X}_j).$$

Next,

$$D_{f,n}^L(X, Y) - D_f^L(X, Y) = (\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}.$$

Since $\mathbb{E}_{P_0}\{Y - \tilde{f}(X)\} = 0$, it follows that

$$\begin{aligned}
\frac{1}{M}\sum_{i=1}^{n}\{D_{f,n}^L(X_i, Y_i) - D_f^L(X_i, Y_i)\} &= \rho_n(\mathbb{P}_n^L - P_0)\Big[(\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}\Big] \\
&= \left(1 - \frac{\rho_n}{\rho_0}\right)(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\}.
\end{aligned}$$

Because Y and $f(X)$ have finite second moments, $\tilde{f}(X)$ also has finite second moment, so

$$(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Since $\rho_n \to \rho_0 \in (0, 1)$ by Assumption 1,

$$\frac{1}{M}\sum_{i=1}^{n}\{D_{f,n}^L(X_i, Y_i) - D_f^L(X_i, Y_i)\} = o_p(M^{-1/2}).$$

Therefore

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D_f^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D_f^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

which proves the stated asymptotic linearity. The mean-zero property of D_f^L and D_f^U follows from

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}\{\tilde{f}(X)\} = \psi_0.$$

For the asymptotic variance, write

$$\delta(X) := \tilde{f}(X) - \mu_0(X), \qquad \varepsilon := Y - \mu_0(X).$$

Then

$$\mathbb{E}_{P_0}\{\delta(X)\} = 0, \qquad \mathbb{E}_{P_0}(\varepsilon \mid X) = 0, \qquad \mathbb{E}_{P_0}(\varepsilon^2) = \mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Also,

$$D_f^U(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0 + \delta(\widetilde{X}),$$

and, since $Y - \tilde{f}(X) = \varepsilon - \delta(X)$,

$$\begin{aligned} D_f^L(X,Y) &= \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\} \\ &= \mu_0(X) - \psi_0 + \delta(X) + \rho_0^{-1}\{\varepsilon - \delta(X)\} \\ &= \mu_0(X) - \psi_0 + \rho_0^{-1}\varepsilon + \big(1 - \rho_0^{-1}\big)\delta(X). \end{aligned}$$

Set

$$A(X) := \mu_0(X) - \psi_0, \qquad B(X) := \delta(X), \qquad C(X,Y) := \rho_0^{-1}\varepsilon.$$

Then $\mathbb{E}_{P_0}\{C(X,Y) \mid X\} = 0$, so $C(X,Y)$ is uncorrelated with both $A(X)$ and $B(X)$. Hence

$$\mathrm{Var}\{D_f^L(X,Y)\} = \mathrm{Var}\big\{A(X) + \big(1 - \rho_0^{-1}\big)B(X)\big\} + \mathrm{Var}\{C(X,Y)\},$$

while

$$\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{A(\widetilde{X}) + B(\widetilde{X})\}.$$

Therefore

$$\begin{aligned} &\rho_0\mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} \\ &= \rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) + \rho_0\mathrm{Var}(C). \end{aligned}$$

Expanding the two variances involving A and B gives

$$\begin{aligned} &\rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) \\ &= \{\rho_0 + (1-\rho_0)\}\mathrm{Var}(A) \\ &\quad + \Big\{\rho_0\big(1 - \rho_0^{-1}\big)^2 + (1-\rho_0)\Big\}\,\mathrm{Var}(B) \\ &\quad + 2\left\{\rho_0\big(1 - \rho_0^{-1}\big) + (1-\rho_0)\right\}\mathrm{Cov}(A,B). \end{aligned}$$

Now

$$1 - \rho_0^{-1} = -\frac{1-\rho_0}{\rho_0},$$

so

$$\rho_0\big(1 - \rho_0^{-1}\big)^2 + (1-\rho_0) = \frac{(1-\rho_0)^2}{\rho_0} + (1-\rho_0) = \frac{1-\rho_0}{\rho_0},$$

and

$$\rho_0\big(1 - \rho_0^{-1}\big) + (1-\rho_0) = -(1-\rho_0) + (1-\rho_0) = 0.$$

Thus

$$\rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) = \mathrm{Var}(A) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}(B).$$

Also,

$$\rho_0\mathrm{Var}(C) = \rho_0 \cdot \rho_0^{-2}\mathbb{E}_{P_0}(\varepsilon^2) = \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Since $\mathbb{E}_{P_0}\{B(X)\} = 0$,

$$\mathrm{Var}(B) = \mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

Combining the preceding displays yields

$$\rho_0\mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

This expression is minimized when $\tilde{f}(X) = \mu_0(X)$ almost surely, in particular at $f = \mu_0$. Standard semiparametric characterization results for missing-data models imply that this minimum is the semiparametric efficiency bound in the unrestricted two-sample model; see, for example, Bickel et al. [10], Robins et al. [82], Robins and Rotnitzky [80], van der Laan and Robins [114], Tsiatis [101]. This completes the proof. □

Lemma 2 (Bounded variation after isotonic post-processing)**.** *Suppose Assumption 3 holds. Let $T := m(X)$ and $S_n := m^\star_{n,\mathrm{iso}}(X)$. Define*

$$\eta(t) := \mathbb{E}_{P_0}[Y \mid T = t], \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s].$$

If $t \mapsto \eta(t)$ has total variation at most V, then, conditional on the labeled sample, the map $s \mapsto \eta_n(s)$ has total variation at most V. Moreover, $|\eta_n(s)| \le C_0$ on the support of S_n.

Proof. We adapt the proof of Lemma 6 in [106]. Let $S_n = f_n(T)$, where f_n is the fitted isotonic map. Conditional on the labeled sample, f_n is nondecreasing and piecewise constant. Hence, for each s in the support of S_n, there exists an interval $B_s \subset \mathbb{R}$ such that

$$\{S_n = s\} = \{T \in B_s\}.$$

Therefore

$$\eta_n(s) = \mathbb{E}_{P_0}[Y \mid S_n = s] = \mathbb{E}_{P_0}[\eta(T) \mid T \in B_s].$$

Let $\eta = \eta^+ - \eta^-$ be the Jordan decomposition of η, where η^+ and η^- are nondecreasing and

$$\|\eta\|_{TV} = \|\eta^+\|_{TV} + \|\eta^-\|_{TV} \le V.$$

Then

$$\eta_n(s) = \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] - \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s].$$

If $s_1 < s_2$, then monotonicity of f_n implies every point of B_{s_1} is no larger than every point of B_{s_2}. Hence the conditional distribution of $T \mid T \in B_{s_2}$ first-order stochastically dominates that of $T \mid T \in B_{s_1}$. Therefore, for any nondecreasing function g,

$$\mathbb{E}_{P_0}[g(T) \mid T \in B_{s_1}] \le \mathbb{E}_{P_0}[g(T) \mid T \in B_{s_2}].$$

Applying this with $g = \eta^+$ and $g = \eta^-$, both maps

$$s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \qquad \text{and} \qquad s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]$$

are nondecreasing.

Moreover, each of these conditional expectations takes values in the range of the corresponding function, so

$$\big\|s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^+\|_{TV}, \qquad \big\|s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^-\|_{TV}.$$

Hence

$$\|\eta_n\|_{TV} \le \|\eta^+\|_{TV} + \|\eta^-\|_{TV} = \|\eta\|_{TV} \le V.$$

Finally, if $|\eta(t)| \le C_0$ on the support of T, then

$$|\eta_n(s)| = |\mathbb{E}_{P_0}[\eta(T) \mid T \in B_s]| \le C_0$$

for all s in the support of S_n. □

Lemma 3 (Empirical calibration centers the limit)**.** *Suppose Assumptions 1 and 2 hold and the calibration score equations contain the constant score $h \equiv 1$. Then*

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

Proof. Let $T := m(X)$, and write $m_0 = f_0(T)$, where

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\mathrm{iso}}} \mathbb{E}_{P_0}\big[(Y - f(T))^2\big].$$

Because $\mathcal{F}_{\mathrm{iso}}$ is closed under addition of constants, for every $c \in \mathbb{R}$ the function $f_0 + c$ also belongs to $\mathcal{F}_{\mathrm{iso}}$. Hence the function

$$g(c) := \mathbb{E}_{P_0}\big[(Y - f_0(T) - c)^2\big]$$

is minimized at $c = 0$. Since g is differentiable,

$$0 = g'(0) = -2\,\mathbb{E}_{P_0}[Y - f_0(T)].$$

Therefore

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}[f_0(T)].$$

Recalling that $m_0(X) = f_0(m(X)) = f_0(T)$ and $\psi_0 = \mathbb{E}_{P_0}[Y]$, we conclude

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

□

Proof convention. In the following proofs, fix $\delta > 0$. By $\sup_x |m^\star_{n,\mathrm{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\delta \geq C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \left\{ \sup_x |m^\star_{n,\mathrm{iso}}(x)| \leq C \right\}$$

satisfies $P_0(A_{n,C}) \geq 1 - \delta$ for all $n \geq n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since δ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 4 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof of Lemma 4. Let $T := m(X)$ and let

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\mathrm{iso}},\ \|f(T)\|_\infty \leq C\}.$$

By Assumption 2, $g_0(T)$ is bounded by $C_0 \leq C$, and on $A_{n,C}$ we have $m^\star_{n,\mathrm{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $g_0(T)$, the population isotonic projection m_0 also belongs to $\mathcal{M}_C$. Thus the problem is a one-dimensional isotonic least-squares regression of Y on the score T. Let

$$g_0(T) := \mathbb{E}_{P_0}[Y \mid T].$$

Then, for every $m \in \mathcal{M}_C$,

$$P_0\{(Y - m)^2\} = P_0\{(Y - g_0)^2\} + P_0\{(g_0 - m)^2\},$$

which shows that m_0 is the $L^2(P_T)$-projection of g_0 onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - m_0 :\ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - m_0(T) - g(T))^2 - (Y - m_0(T))^2$, where $Z = (X, Y)$. Since $m^\star_{n,\mathrm{iso}}$ minimizes the empirical squared risk over $\mathcal{F}_{\mathrm{iso}}$, on $A_{n,C}$ the random element $\hat{g} := m^\star_{n,\mathrm{iso}} - m_0$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}^L_n \ell_{\hat{g}} \leq 0 = \mathbb{P}^L_n \ell_0.$$

Moreover,

$$P_0 \ell_g = \|g\|^2_{2,P_{0,X}} - 2P_0\left[(Y - m_0)g\right].$$

Because g is measurable with respect to T, the tower property gives

$$P_0\left[(Y - m_0)g\right] = P_0\left[(g_0 - m_0)g\right].$$

Since m_0 is the $L^2(P_T)$-projection of g_0 onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$P_0\left[(g_0 - m_0)(m - m_0)\right] \leq 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$P_0 \ell_g \geq \|g\|^2_{2,P_{0,X}},$$

so squared loss has the required quadratic margin around m_0.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \leq \delta\}.$$

Because every $m \in \mathcal{M}_C$ is bounded by C, every $g \in \mathcal{F}_\delta$ is uniformly bounded, and the standard bracketing bound for one-dimensional monotone classes gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta/\varepsilon, \qquad 0 < \varepsilon < \delta.$$

Therefore

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner [118].

Write $\xi := Y - m_0(T)$, so that $\ell_g = g^2 - 2\xi g$. The class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$, so by Lemma 3.4.2 of van der Vaart and Wellner [118] and the same entropy bound,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\, |(\mathbb{P}_n^L - P_0)(g^2)| \lesssim \delta^{1/2}.$$

For the multiplier term, Assumption 2 implies that $Y - g_0(T)$ is sub-Gaussian or subexponential. Since both $g_0(T)$ and $m_0(T)$ are bounded by C, the difference $g_0(T) - m_0(T)$ is bounded, and therefore $\xi = Y - m_0(T)$ is also sub-Gaussian or subexponential. Thus the product class $\xi\mathcal{F}_\delta := \{\xi g : g \in \mathcal{F}_\delta\}$ satisfies the same localized bracketing bound in the Bernstein-type norm used in the subexponential least-squares proof of Bibaut and van der Laan [9], yielding

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\, |(\mathbb{P}_n^L - P_0)(\xi g)| \lesssim \delta^{1/2}.$$

Combining the previous two displays,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\, |(\mathbb{P}_n^L - P_0)\ell_g| \lesssim \delta^{1/2}.$$

We may therefore apply Theorem 3.4.1 of van der Vaart and Wellner [118] with $d(g) = \|g\|_{2,P_{0,X}}$, quadratic margin $P_0\ell_g \gtrsim d^2(g)$, and modulus $\phi_n(\delta) \asymp \delta^{1/2}$. The fixed-point condition $r_n^2\phi_n(r_n^{-1}) \lesssim \sqrt{n}$ gives $r_n \asymp n^{1/3}$, and hence

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

The preceding display is obtained on $A_{n,C}$. Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same rate holds unconditionally. This is the classical one-dimensional isotonic rate, stated directly in the random-design $L^2(P_{0,X})$ norm relevant for the theorem. □

Theorem 4 (Calibration error of isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X)].$$

Then

$$\|\widehat{m}_0 - m_{n,\mathrm{iso}}^\star\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Proof. The proof follows that of Theorem 1 in [106]; see also [102]. Let

$$T := m(X), \qquad S_n := m_{n,\mathrm{iso}}^\star(X), \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s],$$

and define

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\mathrm{iso}}^\star(X) = \eta_n(S_n) - S_n.$$

Conditional on the labeled sample, $h_n(s) := \eta_n(s) - s$ is a measurable function of s. Applying proposition 1 pathwise with $h = h_n$ yields

$$\mathbb{P}_n^L[\Delta_n(X)\{Y - S_n\}] = 0.$$

Also,

$$P_0[\Delta_n(X)\{Y - S_n\}] = P_0[\Delta_n(X)\,\mathbb{E}_{P_0}[Y - S_n \mid S_n]] = P_0(\Delta_n^2) = \|\Delta_n\|_{2,P_{0,X}}^2.$$

Hence

$$\|\Delta_n\|_{2,P_{0,X}}^2 = (P_0 - \mathbb{P}_n^L)[\Delta_n(X)\{Y - S_n\}]\,.$$

Fix $\delta > 0$, and work on the event $A_{n,C}$ from the proof convention. On this event, S_n is uniformly bounded by C. By Lemma 2, the function η_n belongs to the bounded-variation class

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$

where $V < \infty$ is the variation bound in Assumption 3. Let

$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

Since $S_n = f_n(T)$ for a nondecreasing fitted isotonic map f_n, and composition with a monotone map preserves bounded variation, the function

$$t \mapsto \eta_n\{f_n(t)\}$$

also belongs to $\mathcal{F}_{\mathrm{TV},C}$, after enlarging C if necessary. Thus $\eta_n(S_n) = h_1(T)$ for some $h_1 = f_1 \circ T$ with $f_1 \in \mathcal{F}_{\mathrm{TV},C}$, while $S_n = h_2(T)$ for $h_2 = f_2 \circ T$ with $f_2 = f_n \in \mathcal{F}_{\mathrm{iso},C}$.

Define the localized product class

$$\mathcal{G}_{n,C,\delta} := \Big\{(h_1 - h_2)(Y - h_2) : h_1 = f_1 \circ T,\ h_2 = f_2 \circ T,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le \delta\Big\}.$$

Then

$$\Delta_n(X)\{Y - S_n\} \in \mathcal{G}_{n,C,\delta_n}, \qquad \delta_n := \|\Delta_n\|_{2,P_{0,X}}.$$

Because bounded-variation functions are differences of bounded monotone functions, standard entropy-preservation results for Lipschitz transformations imply that the localized entropy integral of $\mathcal{G}_{n,C,\delta}$ satisfies

$$J_{[]}(\delta, \mathcal{G}_{n,C,\delta}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Moreover, Assumption 2 implies that $Y - \eta(T)$ is sub-Gaussian or subexponential, and on $A_{n,C}$ the difference $\eta(T) - S_n$ is bounded. Therefore $Y - S_n$ has the same tail type, up to constants. The same localized multiplier empirical-process argument used in the proof of Lemma 4 then gives

$$E^* \sup_{g \in \mathcal{G}_{n,C,\delta}} |(\mathbb{P}_n^L - P_0)g| \lesssim n^{-1/2}\delta^{1/2}.$$

Set $\varepsilon_n := n^{-1/3}$. For $s \ge 0$, let

$$B_s := \{2^s \varepsilon_n \le \delta_n < 2^{s+1}\varepsilon_n\}.$$

On $B_s \cap A_{n,C}$,

$$2^{2s}\varepsilon_n^2 \le \delta_n^2 \le \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|.$$

Therefore, by Markov's inequality,

$$P_0(B_s \cap A_{n,C}) \lesssim \frac{E^* \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|}{2^{2s}\varepsilon_n^2} \lesssim \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \lesssim 2^{-3s/2},$$

since $n^{-1/2}\varepsilon_n^{-3/2} = 1$. Summing over $s \ge S$ yields

$$P_0(\delta_n \ge 2^S \varepsilon_n,\ A_{n,C}) \lesssim \sum_{s=S}^{\infty} 2^{-3s/2} \xrightarrow[S \to \infty]{} 0.$$

Because $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, it follows that

$$\|\Delta_n\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

This is exactly the claimed calibration-error rate. □

Proof of Lemma 3. Because $h \equiv 1$ belongs to the calibration score class, equation (5) gives

$$\mathbb{P}_n^L\{Y - m_{n,\mathrm{iso}}^\star(X)\} = 0.$$

Hence

$$\mathbb{P}_n^L Y - \mathbb{P}_n^L m_0(X) = \mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\}.$$

By Cauchy–Schwarz and Lemma 4,

$$P_{0,X}\left|m_{n,\mathrm{iso}}^\star - m_0\right| \le \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}),$$

and Lemma 5 gives

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Therefore

$$\mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = P_{0,X}\{m_{n,\mathrm{iso}}^\star - m_0\} + (\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = o_p(1).$$

By Assumption 1, the law of large numbers yields $\mathbb{P}_n^L Y \to_p \psi_0$ and $\mathbb{P}_n^L m_0(X) \to_p \mathbb{E}_{P_{0,X}}[m_0(X)]$. Therefore $\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0$. □

Lemma 5 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star(\widetilde{X}) - m_0(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 5. By Lemma 4,

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

To control the centered empirical-process term, work on the event $A_{n,C}$ from the proof convention and consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - m_0 : \ f \in \mathcal{F}_{\mathrm{iso}}, \ \|f \circ m\|_\infty \le C, \ \|g\|_{2,P_{0,X}} \lesssim n^{-1/3}\}.$$

On $A_{n,C}$, the function $m_{n,\mathrm{iso}}^\star - m_0$ belongs to $\mathcal{F}_{n,C}$. Because m is fixed, this is a localized class of bounded monotone functions of a one-dimensional score. Standard bracketing entropy bounds for monotone classes imply

$$J_{[]}(r, \mathcal{F}_{n,C}, L_2(P_0)) \lesssim r^{1/2}.$$

Applying a local maximal inequality for empirical processes, such as Lemma 3.4.2 of van der Vaart and Wellner [118], at radius $r_n \asymp n^{-1/3}$ yields

$$\sup_{g \in \mathcal{F}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})g| = O_p\Big(n^{-1/2} r_n^{1/2}\Big) = O_p(n^{-2/3}).$$

In particular,

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, $m_{n,\mathrm{iso}}^\star - m_0$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p\left(N^{-1/2}\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 4. □

Lemma 6 (Centered empirical-process bounds for isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\mathrm{iso}}^\star(X).$$

Then

$$(\mathbb{P}_n^L - P_{0,X})\Delta_n(X) = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof. By theorem 4,

$$\|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Let $V < \infty$ be the variation bound in Assumption 3. Fix $\delta > 0$, and work on the event $A_{n,C}$. On this event, $m_{n,\mathrm{iso}}^\star$ is uniformly bounded by C. By Lemma 2, the function

$$s \mapsto \eta_n(s) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X) = s]$$

has total variation at most V and is bounded by C on the support of $m_{n,\mathrm{iso}}^\star(X)$. Since $m_{n,\mathrm{iso}}^\star(X) = f_n(T)$ with $T = m(X)$ and f_n nondecreasing, it follows that

$$\widehat{m}_0(X) = \eta_n\{f_n(T)\}$$

is a bounded-variation transform of T, while $m_{n,\mathrm{iso}}^\star(X)$ is a bounded monotone transform of T.

Consider the classes

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C, \ \|f\|_{TV} \le V\},$$
$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\},$$

and define

$$\mathcal{H}_{n,C} := \Big\{h_1 - h_2 : h_1 = f_1 \circ T, \ h_2 = f_2 \circ T, \ f_1 \in \mathcal{F}_{\mathrm{TV},C}, \ f_2 \in \mathcal{F}_{\mathrm{iso},C}, \ \|h_1 - h_2\|_{2,P_{0,X}}^2 \lesssim n^{-2/3}\Big\}.$$

On $A_{n,C}$, the random function Δ_n belongs to $\mathcal{H}_{n,C}$.

Because bounded-variation functions are differences of bounded monotone functions, the localized bracketing entropy of $\mathcal{H}_{n,C}$ obeys the same bound as in the proof of theorem 4, namely

$$J_{[]}(\delta, \mathcal{H}_{n,C}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Applying the same local maximal inequality at localization radius $n^{-1/3}$ therefore gives, on $A_{n,C}$,

$$|(\mathbb{P}_n^L - P_{0,X})\Delta_n(X)| = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \leq \delta$ for all large n, and $\delta > 0$ was arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, Δ_n is fixed and the unlabeled sample is independent. Hence

$$(\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\left(N^{-1/2}\|\Delta_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by theorem 4. □

J Proof of asymptotic linearity

Proof of theorem 2. The proof combines the exact AIPW representation from Theorem 1 with a localized empirical-process bound for the isotonic calibrator class. The latter controls the remainder generated by estimating the monotone post-processing map and yields the $n^{-2/3}$ remainder rate. The overall argument is closely related to the calibrated DML framework of van der Laan et al. [108]; see also Kennedy [54] for background on semiparametric influence-function arguments.

By Lemma 3, $P_{0,X}m_0 = \psi_0$. Using theorem 1 with $\widehat{w} \equiv 1$,

$$\widehat{\psi}_{\text{iso}} = \rho_n\mathbb{P}_n^L\{m_{n,\text{iso}}^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_{n,\text{iso}}^\star(\widetilde{X})\} + \rho_n\mathbb{P}_n^L\{Y - m_{n,\text{iso}}^\star(X)\}.$$

Add and subtract the AIPW form built from m_0:

$$\widehat{\psi}_{\text{iso}} = \rho_n\mathbb{P}_n^L\left\{m_0 + \rho_n^{-1}(Y - m_0)\right\} + (1-\rho_n)\mathbb{P}_N^U\{m_0(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\}.$$

Subtracting ψ_0 gives

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_0 - \psi_0\} \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\}.\end{aligned}$$

To replace ρ_n^{-1} by ρ_0^{-1}, note that

$$\rho_n(\mathbb{P}_n^L - P_0)\Big\{(\rho_n^{-1} - \rho_0^{-1})(Y - m_0)\Big\} = \Big(1 - \frac{\rho_n}{\rho_0}\Big)(\mathbb{P}_n^L - P_0)(Y - m_0).$$

The right-hand side is $o_p(M^{-1/2})$ because $\rho_n \to \rho_0$ and

$$(\mathbb{P}_n^L - P_0)(Y - m_0) = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Also,

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_{n,\text{iso}}^\star - m_0\} - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{m_{n,\text{iso}}^\star - m_0\}.$$

By Lemma 5, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Thus the remainder $R_{n,N}$, defined as the difference between $\widehat{\psi}_{\text{iso}} - \psi_0$ and the leading empirical-process terms in equation (14), satisfies the bound stated in theorem 2. Since $n^{-2/3} = o(M^{-1/2})$ and $n^{-1/3}N^{-1/2} = o(M^{-1/2})$, this yields equation (14). □

Proof of corollary 1. By theorem 2,

$$\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{M}}\sum_{i=1}^{n} D_{m_0}^L(X_i, Y_i) + \frac{1}{\sqrt{M}}\sum_{j=1}^{N} D_{m_0}^U(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent, each summand has mean zero, and $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1. A standard triangular-array central limit theorem therefore yields the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := m^\star_{n,\mathrm{iso}} - m_0$. By Lemma 4,

$$P_{0,X}(g_n^2) = \|g_n\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Moreover, the proof of Lemma 4 already established the localized empirical-process bound

$$\sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}_n^L - P_{0,X})(g^2)| = O_p(n^{-1/2}\delta^{1/2}),$$

for the squared class $\{g^2 : g \in \mathcal{F}_\delta\}$. Taking $\delta \asymp n^{-1/3}$, which matches the localization radius of g_n, gives

$$(\mathbb{P}_n^L - P_{0,X})(g_n^2) = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}_n^L(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}_N^U - P_{0,X})(g_n^2) = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Therefore

$$\mathbb{P}_N^U(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n and arbitrary $\delta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\mathrm{iso}} \to_p \psi_0$ follows from theorem 2.

Finally,

$$\widehat{D}^L - D^L_{m_0} = (1-\rho_n^{-1})g_n - (\widehat{\psi}_{\mathrm{iso}} - \psi_0) + (\rho_n^{-1} - \rho_0^{-1})(Y - m_0),$$

and

$$\widehat{D}^U - D^U_{m_0} = g_n(\widetilde{X}) - (\widehat{\psi}_{\mathrm{iso}} - \psi_0).$$

Because $\rho_n \to \rho_0$, $Y - m_0$ has finite second moment, and $\mathbb{P}_n^L(g_n^2), \mathbb{P}_N^U(g_n^2) = o_p(1)$, it follows that

$$\mathbb{P}_n^L\Big[\{\widehat{D}^L - D^L_{m_0}\}^2\Big] = o_p(1), \qquad \mathbb{P}_N^U\Big[\{\widehat{D}^U - D^U_{m_0}\}^2\Big] = o_p(1).$$

Therefore $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

K Proof of post-processing equivalence and reduced-model efficiency

Proof of proposition 4. Let

$$\Delta_n(X) := \widehat{m}_0(X) - m^\star_{n,\mathrm{iso}}(X).$$

Using the definition of the AIPW class together with $\mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{iso}}(X)\} = 0$, we have

$$\begin{aligned}
\widehat{\psi}_{\mathrm{eff}} - \widehat{\psi}_{\mathrm{iso}} &= \rho_n \mathbb{P}_n^L \Delta_n(X) + (1-\rho_n)\mathbb{P}_N^U \Delta_n(\widetilde{X}) + \mathbb{P}_n^L\{Y - \widehat{m}_0(X)\} - \mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{iso}}(X)\} \\
&= \rho_n \mathbb{P}_n^L \Delta_n(X) + (1-\rho_n)\mathbb{P}_N^U \Delta_n(\widetilde{X}) - \mathbb{P}_n^L \Delta_n(X) \\
&= (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n.
\end{aligned}$$

Now decompose

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\Delta_n - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\Delta_n.$$

By Lemma 6, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Since $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1, both rates are $o_p(M^{-1/2})$. Therefore

$$\widehat{\psi}_{\mathrm{eff}} - \widehat{\psi}_{\mathrm{iso}} = (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n = o_p(M^{-1/2}).$$

This proves the claimed first-order equivalence. In particular, $\widehat{\psi}_{\mathrm{eff}}$ and $\widehat{\psi}_{\mathrm{iso}}$ have the same asymptotic variance. □

Proof of corollary 2. For part (i), if $m_0 = \mu_0$ almost surely, then the asymptotic influence pair in theorem 2 is

$$D^L_{\mu_0}(X, Y) = \mu_0(X) - \psi_0 + \rho_0^{-1}\{Y - \mu_0(X)\}, \qquad D^U_{\mu_0}(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0.$$

By proposition 5, this is the efficient influence pair in the unrestricted two-sample model. Therefore $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

For part (ii), let

$$S := m(X), \qquad \eta_0(s) := \mathbb{E}_{P_0}[Y \mid S = s].$$

proposition 5 applied to the reduced two-sample experiment with scalar covariate S shows that the efficient influence pair in that reduced experiment is

$$D^L_{\eta_0}(S, Y) = \eta_0(S) - \psi_0 + \rho_0^{-1}\{Y - \eta_0(S)\}, \qquad D^U_{\eta_0}(\widetilde{S}) = \eta_0(\widetilde{S}) - \psi_0.$$

If η_0 is monotone increasing, then η_0 belongs to the isotonic class and

$$\mathbb{E}_{P_0}\left[\{Y - g(S)\}^2\right] = \mathbb{E}_{P_0}\left[\{Y - \eta_0(S)\}^2\right] + \mathbb{E}_{P_0}\left[\{\eta_0(S) - g(S)\}^2\right]$$

for every measurable g. Hence the population isotonic regression of Y on the score $S = m(X)$ is exactly $\eta_0(S)$, so

$$m_0(X) = \eta_0\{m(X)\} \qquad \text{almost surely.}$$

Therefore, the influence pair in theorem 2 coincides with the reduced-model efficient influence pair above, so $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the reduced experiment indexed by $m(X)$. □

L Pooled i.i.d. Missing-Data Formulation

The joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0, 1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_M(m_n^\star) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - m_n^\star\}\right]$$

up to the same calibration identity as in the main text. The corresponding observed-data efficient influence function is

$$m_0(X) - \psi_0 + \frac{R}{\rho_0}\{Y - m_0(X)\},$$

which is the single-sample analogue of the efficient influence pair obtained by applying proposition 5 to the reduced model with scalar covariate $m(X)$. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

M Additional empirical details

This appendix records the benchmark construction used in section 5.2. We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$. The `galaxies` and semisupervised `census_income` benchmarks use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. In the current draft we rerandomize this split 500 times at each n, always evaluating all estimators on the same split and taking the full-sample mean as the benchmark truth.

The estimator set mirrors the original PPI comparison wherever possible. In the paper-facing summaries, we report the labeled-only estimator, the imputation benchmark on binary-outcome tasks, PPI computed using the official `ppi_py` implementation, the classical AIPW estimator based on the same prediction score, clipped PPI++, and the unclipped efficiency-maximized benchmark AIPW-EM. We then add five calibration-oriented comparators: linear calibration; smooth monotone spline calibration; Platt scaling on the binary-outcome benchmarks; isotonic calibration with a fixed minimum bin size of 10; and an adaptive `AutoCal` selector that chooses among AIPW, linear calibration,

monotone spline calibration, and isotonic calibration by cross-validated empirical efficiency. For the binary-outcome benchmarks, we also report Venn–Abers shrinkage. Platt scaling and Venn–Abers are omitted on `census_income` because that outcome is not binary. Including both PPI++ and AIPW-EM lets us separate the underlying efficiency-maximization idea from the specific `ppi_py` clipping rule for the tuning coefficient, while figure 3 isolates the fixed-calibration comparison among PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`.

For the fixed calibration-based plug-in estimators, we fit the calibrator on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^{n}$, transform both the labeled and unlabeled scores, and then compute the pooled plug-in mean together with the same semisupervised influence-style Wald interval used throughout the benchmark code. AIPW uses the same score but averages it over the pooled covariate sample before adding the labeled residual correction, whereas PPI uses only the unlabeled score average. In theory this retains less labeled-sample score information, though in the finite-sample benchmark results the empirical variance gap is often small and not uniform across all datasets. PPI++ and AIPW-EM both apply one-dimensional empirical efficiency maximization to the same score, but PPI++ inherits the official `ppi_py` clipping rule $\lambda \in [0, 1]$ whereas AIPW-EM does not. Linear calibration regresses Y on an intercept and the score, then clips the fitted values to the range of predicted outcomes observed in the labeled data to prevent extrapolation on the unlabeled data. `MonoSpline` instead fits a smooth nondecreasing spline of Y on the score, using monotonicity constraints together with a mild roughness penalty, and then applies the same clipped pooled plug-in construction. Platt scaling fits a logistic regression of the binary outcome on a stabilized logit transform of the score, while the fixed-bin isotonic estimator fits a stepwise monotone calibration map and also clips to the labeled outcome range. The `AutoCal` estimator is a selector rather than a new calibrator: in the experiments it compares AIPW, linear calibration, monotone spline calibration, and fixed-bin isotonic calibration using 20-fold cross-validation and the estimated influence-function variance as the selection criterion. To keep this step inexpensive when the unlabeled sample is very large, the foldwise criterion is evaluated on an unlabeled subsample of size $\min(N, 10n)$. The reported summaries are Monte Carlo averages of bias, empirical variance, MSE, interval coverage, and relative efficiency versus PPI across the repeated random splits. The complete numerical outputs, including dataset-level tables and per-metric plots, are generated automatically by the accompanying reproduction pipeline.

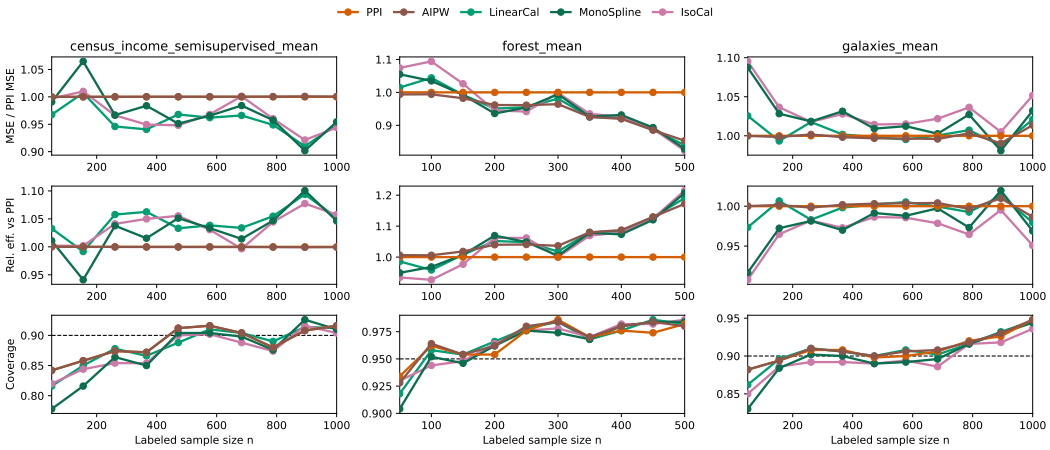

Figure 3: Calibration-focused appendix comparison across the reproduced PPI benchmarks, restricted to PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`. The panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across labeled-sample-size regimes. This figure isolates the fixed-calibration comparison omitted from the main-text benchmark figure for readability.

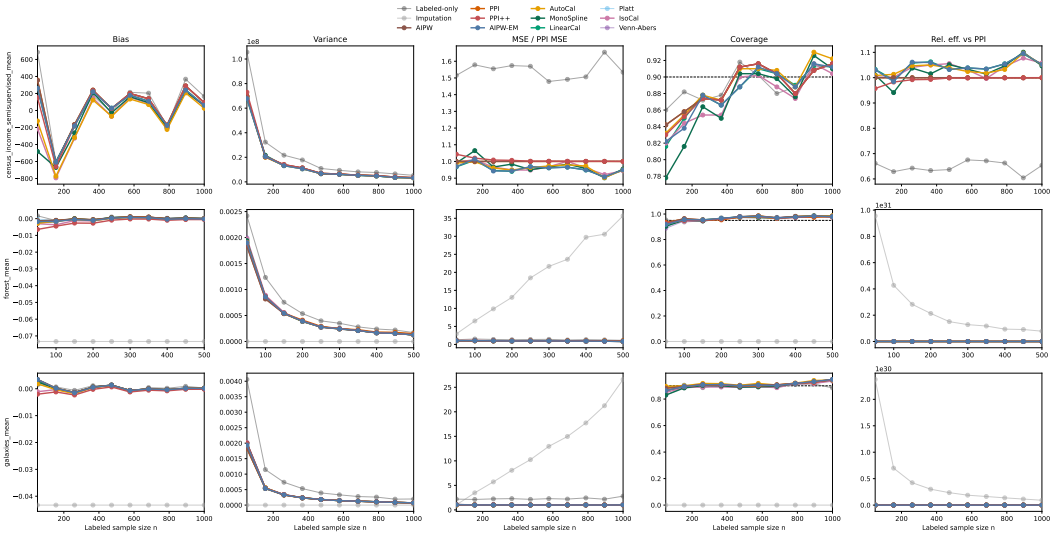

Figure 4: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, normalized MSE relative to PPI, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. In the normalized-MSE and relative-efficiency panels, the dashed horizontal line marks parity with PPI; in the coverage panels it marks the nominal target coverage $1-\alpha$. `AutoCal`, `MonoSpline`, and `IsoCal` appear alongside PPI, AIPW, PPI++, AIPW-EM, and the fixed calibration rules, while Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

Algorithm 1 Isotonic-calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $m^\star_{n,\text{iso}}(X_i) = \hat{f}(T_i)$ and $m^\star_{n,\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}$
6: Set $\widehat{\text{SE}}_{\text{iso}}^2 = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

N `Python` Code

Code availability. To preserve double-blind review, repository and documentation links are omitted from the submission PDF. This appendix instead gives a minimal anonymous implementation sketch that can be bundled with any supplementary artifact.

Below we also record the three main algorithmic summaries from the full paper, using the notation of the main text.

N.1 Algorithmic summaries

Below, we provide a minimal, self-contained implementation of the method introduced in this paper.

N.2 Isotonic regression via XGBoost

```
import numpy as np
import xgboost as xgb
```

Algorithm 2 Linearly calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

Algorithm 3 Linearly calibrated PPI with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (a + X_i^\top \beta + bT_i)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a} + X_i^\top \hat{\beta} + \hat{b}T_i$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a} + \widetilde{X}_j^\top \hat{\beta} + \hat{b}\widetilde{T}_j$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

```
def isoreg_with_xgboost(x, y, max_depth=15, min_child_weight=20, weights=None):
    """
    Fit a monotone calibrator f so that f(x) is nondecreasing in x.

    Parameters
    ----------
    x : array-like, shape (n,) or (n, 1)
        Prediction score used for calibration. In the PPI application this is
        usually the score m(X), so x is typically one-dimensional.
    y : array-like, shape (n,)
        Labeled outcomes.
    max_depth : int, default=15
        Maximum depth for the one-round XGBoost fit.
    min_child_weight : float, default=20
        Minimum total weight in a terminal node.
    weights : array-like, optional
        Optional observation weights.

    Returns
    -------
    predict_fn : callable
        Function mapping new x values to calibrated scores f(x).
    """
    x = np.asarray(x)
    y = np.asarray(y).reshape(-1)
    x = x.reshape(len(y), -1)

    data = xgb.DMatrix(data=x, label=y, weight=weights)
    params = {
        "max_depth": max_depth,
        "min_child_weight": min_child_weight,
        "monotone_constraints": "(" + ",".join(["1"] * x.shape[1]) + ")",
        "eta": 1.0,
        "gamma": 0.0,
```

```
        "lambda": 0.0,
        "objective": "reg:squarederror",
        "verbosity": 0,
    }
    iso_fit = xgb.train(params=params, dtrain=data, num_boost_round=1)

    def predict_fn(x_new):
        x_new = np.asarray(x_new)
        if x_new.ndim == 1:
            x_new = x_new.reshape(-1, 1)
        data_pred = xgb.DMatrix(data=x_new)
        return iso_fit.predict(data_pred)

    return predict_fn
```

N.3 Isotonic-calibrated plug-in

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def isotonic_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
    min_obs=20,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
```

```
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    calibrator = isoreg_with_xgboost(
        x=score_labeled,
        y=Y_labeled,
        min_child_weight=min_obs,
    )

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = isotonic_calibrated_plugin(
#     Y_labeled=Y,
#     X_labeled=X,
#     X_unlabeled=X_tilde,
#     score_fn=m,
# )
```

N.4 Linear calibration

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
```

```
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def linear_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    X_design = np.column_stack([np.ones(n), score_labeled])
    beta_hat, _, _, _ = np.linalg.lstsq(X_design, Y_labeled, rcond=None)

    def calibrator(score):
        score = np.asarray(score).reshape(-1)
        return beta_hat[0] + beta_hat[1] * score

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
```

```
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = linear_calibrated_plugin(
#     Y_labeled=Y,
#     score_labeled=mX,
#     score_unlabeled=mX_tilde,
# )
```

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	356.4905	69836665.4410	69963750.9366	0.842	1.000
census_income_semisupervised_mean	50	380041	AIPW-EM	260.9833	67728426.6042	67796538.8913	0.822	1.031
census_income_semisupervised_mean	50	380041	AutoCal	-123.2501	69242944.1124	69258134.7078	0.832	1.009
census_income_semisupervised_mean	50	380041	Labeled-only	685.3532	105558402.2427	106028111.3137	0.860	0.662
census_income_semisupervised_mean	50	380041	IsoCal	-215.3832	69697400.2434	69743790.1536	0.820	1.002
census_income_semisupervised_mean	50	380041	LinearCal	260.9294	67622601.4006	67690685.5430	0.816	1.033
census_income_semisupervised_mean	50	380041	MonoSpline	-482.0477	69082916.1270	69315286.1508	0.778	1.011
census_income_semisupervised_mean	50	380041	PPI	356.4473	69834865.9486	69961920.5977	0.842	1.000
census_income_semisupervised_mean	50	380041	PPI++	150.8664	72953053.8328	72975814.4980	0.830	0.957
census_income_semisupervised_mean	577	379514	AIPW	199.8136	6353884.5833	6393810.0648	0.916	1.000
census_income_semisupervised_mean	577	379514	AIPW-EM	177.6067	6114918.2974	6146462.4536	0.912	1.039
census_income_semisupervised_mean	577	379514	AutoCal	134.3746	6201029.6459	6219086.1828	0.910	1.024
census_income_semisupervised_mean	577	379514	Labeled-only	213.2465	9407595.6062	9453069.6946	0.902	0.675
census_income_semisupervised_mean	577	379514	IsoCal	155.0503	6162964.7883	6187005.3977	0.902	1.031
census_income_semisupervised_mean	577	379514	LinearCal	179.8005	6116832.9552	6149161.1594	0.910	1.038
census_income_semisupervised_mean	577	379514	MonoSpline	162.1952	6146069.6864	6172376.9541	0.904	1.033
census_income_semisupervised_mean	577	379514	PPI	199.7932	6351836.6166	6391753.9369	0.916	1.000
census_income_semisupervised_mean	577	379514	PPI++	198.7409	6362047.6345	6401545.5819	0.916	0.998
census_income_semisupervised_mean	1000	379091	AIPW	92.1091	3462051.4638	3470535.5425	0.916	0.999
census_income_semisupervised_mean	1000	379091	AIPW-EM	61.4705	3288235.4312	3292014.0588	0.912	1.052
census_income_semisupervised_mean	1000	379091	AutoCal	25.8207	3289467.9320	3290134.6381	0.922	1.052
census_income_semisupervised_mean	1000	379091	Labeled-only	165.4763	5293296.2822	5320678.6872	0.914	0.654
census_income_semisupervised_mean	1000	379091	IsoCal	51.2697	3271622.4413	3274251.0223	0.904	1.058
census_income_semisupervised_mean	1000	379091	LinearCal	60.7501	3287796.5296	3291487.1058	0.912	1.052
census_income_semisupervised_mean	1000	379091	MonoSpline	60.9626	3306575.6859	3310292.1289	0.910	1.046
census_income_semisupervised_mean	1000	379091	PPI	91.9155	3460071.7259	3468520.1894	0.916	1.000
census_income_semisupervised_mean	1000	379091	PPI++	91.9155	3460071.7259	3468520.1894	0.916	1.000
forest_mean	50	1546	AIPW	-0.0012	0.0018	0.0018	0.928	1.006
forest_mean	50	1546	AIPW-EM	-0.0018	0.0019	0.0019	0.918	0.969
forest_mean	50	1546	AutoCal	-0.0028	0.0019	0.0019	0.924	0.965
forest_mean	50	1546	Labeled-only	0.0017	0.0024	0.0024	0.954	0.762
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	9564441057429144180180373209088.000
forest_mean	50	1546	IsoCal	-0.0030	0.0020	0.0020	0.930	0.934
forest_mean	50	1546	LinearCal	-0.0016	0.0019	0.0019	0.918	0.986
forest_mean	50	1546	MonoSpline	-0.0021	0.0019	0.0019	0.904	0.949
forest_mean	50	1546	Platt	-0.0003	0.0019	0.0019	0.900	0.948
forest_mean	50	1546	PPI	-0.0013	0.0018	0.0018	0.934	1.000
forest_mean	50	1546	PPI++	-0.0064	0.0019	0.0019	0.928	0.969
forest_mean	50	1546	Venn-Abers	-0.0027	0.0020	0.0020	0.888	0.923
forest_mean	300	1296	AIPW	0.0011	0.0002	0.0002	0.984	1.036
forest_mean	300	1296	AIPW-EM	0.0007	0.0002	0.0002	0.982	1.012
forest_mean	300	1296	AutoCal	0.0006	0.0002	0.0002	0.978	1.002
forest_mean	300	1296	Labeled-only	0.0007	0.0003	0.0004	0.968	0.706
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	1281441200512508097500322725888.000
forest_mean	300	1296	IsoCal	0.0010	0.0002	0.0002	0.978	1.000
forest_mean	300	1296	LinearCal	0.0010	0.0002	0.0002	0.984	1.018
forest_mean	300	1296	MonoSpline	0.0010	0.0002	0.0002	0.974	1.004
forest_mean	300	1296	Platt	0.0010	0.0002	0.0002	0.982	1.020
forest_mean	300	1296	PPI	0.0011	0.0002	0.0002	0.986	1.000
forest_mean	300	1296	PPI++	-0.0002	0.0002	0.0002	0.980	0.991
forest_mean	300	1296	Venn-Abers	0.0011	0.0002	0.0002	0.978	1.015
forest_mean	500	1096	AIPW	0.0001	0.0001	0.0001	0.980	1.172
forest_mean	500	1096	AIPW-EM	-0.0002	0.0001	0.0001	0.980	1.188
forest_mean	500	1096	AutoCal	-0.0003	0.0001	0.0001	0.986	1.180
forest_mean	500	1096	Labeled-only	0.0001	0.0002	0.0002	0.972	0.879
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	7819091278107866750853379522 56.000
forest_mean	500	1096	IsoCal	-0.0000	0.0001	0.0001	0.986	1.215
forest_mean	500	1096	LinearCal	0.0001	0.0001	0.0001	0.982	1.191
forest_mean	500	1096	MonoSpline	0.0000	0.0001	0.0001	0.984	1.207
forest_mean	500	1096	Platt	0.0001	0.0001	0.0001	0.984	1.203
forest_mean	500	1096	PPI	0.0000	0.0002	0.0002	0.982	1.000
forest_mean	500	1096	PPI++	-0.0006	0.0001	0.0001	0.978	1.179
forest_mean	500	1096	Venn-Abers	-0.0001	0.0001	0.0001	0.978	1.206
galaxies_mean	50	16693	AIPW	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	AIPW-EM	0.0035	0.0019	0.0020	0.864	0.944
galaxies_mean	50	16693	AutoCal	0.0019	0.0018	0.0018	0.896	0.997
galaxies_mean	50	16693	Labeled-only	0.0014	0.0041	0.0041	0.890	0.451
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	2376289106442477113652488962048.000
galaxies_mean	50	16693	IsoCal	-0.0011	0.0020	0.0020	0.850	0.908
galaxies_mean	50	16693	LinearCal	0.0030	0.0019	0.0019	0.862	0.973
galaxies_mean	50	16693	MonoSpline	0.0026	0.0020	0.0020	0.830	0.917
galaxies_mean	50	16693	Platt	0.0033	0.0019	0.0019	0.856	0.966
galaxies_mean	50	16693	PPI	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	PPI++	-0.0020	0.0020	0.0020	0.864	0.914
galaxies_mean	50	16693	Venn-Abers	0.0022	0.0020	0.0020	0.830	0.907
galaxies_mean	577	16166	AIPW	-0.0006	0.0001	0.0001	0.906	1.004
galaxies_mean	577	16166	AIPW-EM	-0.0007	0.0001	0.0001	0.906	1.005
galaxies_mean	577	16166	AutoCal	-0.0008	0.0001	0.0001	0.918	0.989
galaxies_mean	577	16166	Labeled-only	-0.0011	0.0003	0.0003	0.910	0.436
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	1873962492481163727822 15856128.000
galaxies_mean	577	16166	IsoCal	-0.0007	0.0001	0.0001	0.894	0.986
galaxies_mean	577	16166	LinearCal	-0.0007	0.0001	0.0001	0.908	1.005
galaxies_mean	577	16166	MonoSpline	-0.0006	0.0001	0.0001	0.892	0.988
galaxies_mean	577	16166	Platt	-0.0007	0.0001	0.0001	0.896	1.002
galaxies_mean	577	16166	PPI	-0.0006	0.0001	0.0001	0.900	1.000
galaxies_mean	577	16166	PPI++	-0.0012	0.0001	0.0001	0.902	0.988
galaxies_mean	577	16166	Venn-Abers	-0.0006	0.0001	0.0001	0.892	0.986
galaxies_mean	1000	15743	AIPW	0.0002	0.0001	0.0001	0.948	0.987
galaxies_mean	1000	15743	AIPW-EM	0.0002	0.0001	0.0001	0.948	0.980
galaxies_mean	1000	15743	AutoCal	0.0001	0.0001	0.0001	0.944	0.977
galaxies_mean	1000	15743	Labeled-only	0.0003	0.0002	0.0002	0.884	0.357
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	9146652808341511999670190 0800.000
galaxies_mean	1000	15743	IsoCal	0.0001	0.0001	0.0001	0.936	0.951
galaxies_mean	1000	15743	LinearCal	0.0002	0.0001	0.0001	0.946	0.980
galaxies_mean	1000	15743	MonoSpline	0.0002	0.0001	0.0001	0.944	0.969
galaxies_mean	1000	15743	Platt	0.0002	0.0001	0.0001	0.942	0.979
galaxies_mean	1000	15743	PPI	0.0002	0.0001	0.0001	0.948	1.000
galaxies_mean	1000	15743	PPI++	-0.0002	0.0001	0.0001	0.946	0.977
galaxies_mean	1000	15743	Venn-Abers	0.0002	0.0001	0.0001	0.942	0.952

Table 1: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI for the displayed benchmark set, including PPI++, AIPW-EM, `AutoCal`, `MonoSpline`, and `IsoCal`.

NeurIPS Paper Checklist

1. **Claims. Yes.** The abstract and introduction state the paper's main claims, and the paper distinguishes between exact representation results, asymptotic theory, and empirical findings; see Sections 2–5.
2. **Limitations. Yes.** The conclusion and theory sections state the paper's main scope limitations: the strongest guarantees use the known-ρ_n two-sample design and a stable second-stage calibrator; see Section 6 and Appendix B.
3. **Theory, assumptions, and proofs. Yes.** The main assumptions are stated in Sections 2 and 4, with complete proofs collected in Appendices G, H, I, J, and K.
4. **Experimental results. Yes.** The main experimental takeaways appear in Section 5, with benchmark construction and additional diagnostics deferred to Appendix M.
5. **Open access to data and code. No.** This anonymous submission does not include a public repository or packaged artifact; Appendix N provides representative implementation sketches, and benchmark sources are cited in the text.
6. **Experimental setting and details. Yes.** The simulation design, benchmark protocol, estimator set, and evaluation summaries are described in Section 5 and Appendix M.
7. **Experimental statistical significance. Yes.** The experiments report Monte Carlo variability, interval coverage, and efficiency comparisons, and the inference procedures are tied to the Wald theory in Section 4; see also Appendix C.
8. **Experiments compute resources. No.** The paper does not currently report wall-clock time, hardware, or aggregate compute usage for the simulation and benchmark runs.
9. **Code of ethics. Yes.** The paper is a methodological contribution in semisupervised statistical inference, and we are not aware of any aspect of the work that conflicts with the NeurIPS Code of Ethics.
10. **Broader impacts. N/A.** The paper develops estimation and inference methodology rather than a deployed decision system, though misuse of semisupervised estimators in high-stakes applications remains a general concern.
11. **Safeguards. N/A.** The paper does not introduce a system that generates content or interacts with users, and it does not release new high-risk artifacts requiring special safeguards.
12. **Licenses for existing assets. No.** The current draft cites the benchmark sources and related software ecosystems but does not yet include a consolidated license/version table for all reused assets.
13. **New assets. N/A.** The paper does not introduce a new dataset, benchmark, or foundation model.
14. **Crowdsourcing and research with human subjects. N/A.** The paper does not involve human-subject experiments or crowdsourcing.